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Bachelor's Thesis

A CFD-based Workflow for Multi-Objective Optimization of Inlet Geometry Under Flashback and NO_x Emission Constraints

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Abstract

This thesis develops a CFD-based workflow for the optimization of hydrogen burner inlet geometries with respect to nitrogen oxide (NO_x) emissions and flashback risk. The focus is on establishing a computationally efficient and automated methodology that enables the systematic exploration of design trade-offs under fixed operating conditions and constraints imposed by additive manufacturing.

The workflow integrates steady-state Reynolds-Averaged Navier–Stokes (RANS) simulations into a multi-objective optimization framework. A two-stage simulation strategy is employed. In the first step, a non-reactive three-dimensional inlet simulation is used to resolve the upstream mixing field. The resulting outlet data are then mapped onto a two-dimensional reactive outlet simulation, in which hydrogen combustion is modeled using detailed chemical kinetics. This reactive simulation enables the direct evaluation of thermal NO_x formation and flame stabilization behavior. Flashback risk is assessed using a velocity-gradient-based indicator evaluated at near-wall locations, derived from established boundary-layer flashback criteria.

The complete methodology is embedded into an automated optimization loop coupling OpenFOAM, parametric geometry generation, and the Dakota framework. Due to the multi-objective formulation and the presence of discrete design variables, a multi-objective genetic algorithm (MOGA) is employed to approximate the Pareto-optimal set of inlet geometries. The resulting optimization studies demonstrate that the developed workflow is capable of resolving the trade-off between NO_x emissions, flashback risk, and pressure loss within the limitations of steady-state RANS modeling.

The proposed framework provides a modular basis for CFD-driven inlet design and can be extended toward higher-fidelity turbulence modeling and surrogate-assisted optimization strategies in future work.

Zusammenfassung

In dieser Arbeit wird ein CFD-basierter Arbeitsablauf zur Optimierung von Wasserstoffbrenner-Einlassgeometrien hinsichtlich der Stickoxiddbildung (NO_x) und des Flammenrückschlag-Risikos entwickelt. Ziel ist die Etablierung einer recheneffizienten und automatisierten Methodik, die eine systematische Untersuchung von Zielkonflikten unter festen Betriebsbedingungen sowie unter Randbedingungen der additiven Fertigung ermöglicht.

Der entwickelte Arbeitsablauf integriert stationäre Reynolds-Averaged-Navier-Stokes-(RANS)-Simulationen in ein multi-objektives Optimierungsframework. Hierzu wird eine zweistufige Simulationsstrategie eingesetzt. Zunächst wird eine nichtreaktive dreidimensionale Einlasssimulation durchgeführt, um das vorgelagerte Mischungsfeld aufzulösen. Die resultierenden Auslassprofile werden anschließend auf eine zweidimensionale reaktive Auslasssimulation abgebildet, in der die Wasserstoffverbrennung unter Verwendung detaillierter chemischer Kinetik modelliert wird. Diese Simulation erlaubt die direkte Bewertung der thermischen NO_x -Bildung sowie des Flammenstabilisierungsverhaltens. Das Flammenrückschlag-Risiko wird über einen geschwindigkeitsgradientenbasierten Indikator an wandnahen Positionen quantifiziert, der aus etablierten Kriterien für Grenzschicht-Flammenrückschlag abgeleitet ist.

Die vollständige Methodik ist in eine automatisierte Optimierungsschleife eingebettet, welche OpenFOAM, parametrische Geometrierzeugung und das Dakota-Framework koppelt. Aufgrund der multi-objektiven Problemstellung sowie des Vorhandenseins diskreter Entwurfsvariablen wird ein multi-objektiver genetischer Algorithmus (MOGA) zur Approximation der Pareto-optimalen Menge von Einlassgeometrien eingesetzt. Die durchgeführten Optimierungsstudien zeigen, dass der entwickelte Arbeitsablauf in der Lage ist, den Zielkonflikt zwischen NO_x -Emissionen, Flammenrückschlag-Risiko und Druckverlust im Rahmen der Einschränkungen stationärer RANS-Modellierung aufzulösen.

Der vorgestellte Arbeitsablauf stellt eine modulare Grundlage für CFD-gestützte Einlassauslegung dar und kann in zukünftigen Arbeiten um höherwertige Turbulenzmodellierung und surrogate-gestützte Optimierungsstrategien erweitert werden.

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Nomenclature

Formula Symbols and Units		
Symbol	Definition	Unit
p	Pressure	Pa
T	Temperature	K
S_T	Turbulent Flame Speed	m/s
S_f	Local Flame (Burning) Velocity	m/s
s_l	Laminar Flame Speed	m/s
Le	Lewis Number	—
k	Turbulent Kinetic Energy	m ² /s ²
q	Heat Flux	W/m ²
h	Specific enthalpy	J/kg
U	Velocity	m/s
Ze	Zeldovich Number	—
Pe	Péclet Number	—

Greek Formula Symbols and Units		
Symbol	Definition	Unit
ϵ	Turbulent Dissipation Rate	m ² /s ³
ρ	Density	kg/m ³
ν_t	Turbulent Kinematic Viscosity	m ² /s
ω	Specific Dissipation Rate	1/s
α_t	Turbulent Thermal Diffusivity	m ² /s
δ_p	Penetration Distance	m
δ_l	Laminar Flame Thickness	m
ϕ	Equivalence Ratio	—

Indices and Abbreviations

Symbol	Definition
AM	Additive Manufacturing
BLF	Boundary Layer Flashback
CAD	Computer-Aided Design
CDF	Cumulative Distribution Function
CFD	Computational Fluid Dynamics
CIVB	Combustion-induced Vortex Breakdown
CV	Coefficient of Variation
CVG	Critical Velocity Gradient
DNS	Direct Numerical Simulation
EDC	Eddy Dissipation Concept
EGO	Efficient Global Optimization
ET-CVG	Extended Turbulent Critical Velocity Gradient
FRIV	Flow Rate Inlet Velocity
GA	Genetic Algorithm
GAMG	Geometric-Algebraic Multigrid
GP	Gaussian Process
JICF	Jet-in-crossflow
LES	Large Eddy Simulation
LHS	Latin Hypercube Sampling
LOO	Leave-One-Out (Cross Validation Scheme)
LPG	Liquefied Petroleum
MAPE	Mean Absolute Percentage Error
MOGA	Multi-objective Genetic Algorithm
NO _x	Nitrogen Oxides
NSGA-II	Non-dominated Sorting Genetic Algorithm II
OF	OpenFOAM
PAES	Pareto Archived Evolution Strategy
PCG	Preconditioned Conjugate Gradient
PDF	Probability Distribution Function
PIMPLE	PISO–SIMPLE Algorithm
PISO	Pressure Implicit with Splitting of Operators
RANS	Reynolds-Averaged-Navier-Stokes
SIMPLE	Semi-Implicit Method for Pressure-Linked Equations
SLURM	Simple Linux Utility for Resource Management
SPEA2	Strength Pareto Evolutionary Algorithm 2
T-CVG	Turbulent Critical Velocity Gradient
TDIs	Turbulent Diffusion Instabilities
TIKE	Turbulent Intensity Kinetic Energy

continued on next page

Indices and Abbreviations

Symbol	Definition
TMLDR	Turbulent Mixing Length Dissipation Rate
TMLF	Turbulent Mixing Length Frequency

Chapter 1

Introduction

On July 26, 1943, the city of Los Angeles, USA experienced its first major smog episode, during which the weather became yellowish-brownish color, visibility dropped to a few hundred meters and residents reported eye irritation and breathing difficulties [1]. This phenomenon, called photochemical smog or also commonly known as “Los Angeles smog” [2], can reach severe levels within the Los Angeles basin [3], and is composed of human-generated pollutants, including ozone, nitric acid and a range of organic compounds. These unpleasant smelling compounds are confined close to the ground by persistent temperature inversions, and can cause harm to the human health [4].

Photochemical smog is primarily initiated by nitrogen oxides (NO_x). These pollutants are released commonly from internal combustion engines and industry burners [4], which contributed significantly to the frequent photochemical smog episodes observed in the Los Angeles Basin during the early 20th century. At that time, extensive car-dependent commuting resulted in high emissions of NO_x and hydrocarbons [3], while effective emission control technologies were not yet in place [5]. Some emission regulations were implemented in Los Angeles for motor vehicles during the 1960s. However, the first three-way catalytic converters did not appear until 1981, which were designed to control the NO_x emissions [5].

Another possible source of NO_x is hydrogen combustion. The transition toward hydrogen as a combustion fuel has gained significant momentum due to its potential to reduce carbon emissions and mitigate climate change. Unlike hydrocarbon fuels, hydrogen combustion produces no carbon dioxide, which makes it an attractive option for achieving carbon neutrality in energy and industrial applications. However, when hydrogen is burned at high temperatures with air, thermal NO_x is formed, as nitrogen and oxygen from the ambient air react at elevated temperatures. The inherently high flame temperature and rapid reaction kinetics of hydrogen–air mixtures intensify this effect, leading to potentially higher NO_x emissions compared to conventional natural gas combustion [6, 7].

Hong Kong has also dealt with severe photochemical smog at the end of the twentieth century. To mitigate it, a large portion of its taxis and buses were converted to run on liquefied petroleum gas (LPG) between 1997 and 2008, as LPG is a cleaner fuel that produces less NO_x . However, despite the conversion, atmospheric monitoring from 2008 to 2012 unexpectedly showed an increase in ambient NO_x concentrations. This rise was attributed to the deterioration and clogging of catalytic converters in the LPG vehicles, suggesting that inadequate maintenance and delayed replacement of these converters contributed to the problem [8, 9]. The case of Hong Kong illustrates that emission reductions cannot rely solely on fuel substitution or post-combustion treatment technologies such as catalytic converters. Even with the transition to LPG, inadequate combustion design and catalyst degradation led to a measurable increase in NO_x emissions. This emphasizes the necessity of addressing pollutant formation at the source,

through optimally designing more stable and low-emission combustion processes.

However, achieving low NO_x emissions is only one of several factors that must be considered in the optimal design of combustion systems. Another critical phenomenon is flashback, which occurs when the flame propagates upstream from the combustion zone into the premixing section of the combustor. Such an event can lead to severe thermal damage, as the flame causes a substantial temperature rise within the premixer [10]. The likelihood of flashback increases when homogeneous fuel-air mixing is not achieved over a sufficiently short mixing length. This issue is particularly critical for lean premixed or partially premixed combustion systems, which are commonly employed to reduce NO_x emissions [6].

The primary objective of this work is to develop a workflow that integrates Reynolds-Averaged Navier-Stokes (RANS) simulations into an automated optimization framework for identifying an additive manufacturing (AM) inlet geometry that offers the optimal compromise between minimizing flashback risk and reducing NO_x emissions for a given burner power. The overall goal is to achieve a well-mixed, stable flame characterized by low NO_x formation and reduced flashback tendency.

To achieve this objective, NO_x prediction and flashback detection methodologies are developed and implemented as post-processing steps applied to RANS simulations within the OpenFOAM environment, including the identification and incorporation of additional physical models where appropriate. Subsequently, the RANS solver is embedded into an optimization framework that extends an existing baseline workflow. The resulting integrated workflow is then demonstrated through a DAP inlet geometry optimization case study, which serves as a proof of concept for the coupling of RANS-based reactive flow simulations with multi-objective optimization.

Implementing flashback and NO_x detection mechanisms within RANS and embedding them in an automated optimization loop transforms CFD from a diagnostic tool into a design driver and represents a multi-disciplinary approach that bridges turbulent combustion modeling and design optimization. It enables a paradigm shift toward design-by-simulation rather than analysis-by-simulation, and provides a quantitative framework for addressing physical trade-offs between safety (flashback) and sustainability (NO_x emissions).

Furthermore, the flexibility of additive manufacturing makes such geometry optimization industrially meaningful. Additive manufacturing allows for the fabrication of complex inlet geometries that were previously unattainable using conventional methods. A unified design optimization framework that combines DAP and CFD enables more efficient exploration of the design space, compared to conventional manual iterations, and facilitates systematic identification of globally optimal geometries [6].

The primary challenge in this integration lies in the inherent difference between the time-averaged nature of RANS and the transient character of flashback phenomena. In design optimization, RANS simulations remain the preferred choice, as LES and DNS are computationally too expensive for iterative optimization loops [11]. Consequently, the workflow proposed in this work utilizes RANS, despite the model's limitations.

In essence, this work aims to establish a computationally efficient and industrially relevant methodology that contributes to decarbonization while providing safe burner designs. In Chapter 2 the theoretical background is presented and the fundamental concepts are introduced. In Chapter 3 an overview of the current conventional practices is provided. In Chapter 4 the methodology of the proposed workflow is presented. In Chapter 5 the results of the test cases are interpreted and discussed. Finally, in Chapter 6 the study is concluded by summarizing the main insights and proposing potential directions for future research.

Chapter 2

Theoretical Background

2.1 Combustion Fundamentals

Combustion refers to a chemical reaction between a fuel and an oxidizer that releases heat and light [12]. It can be classified into three main types depending on how well the fuel and oxidizer are mixed before the reaction: premixed, nonpremixed, and partially premixed combustion.

In premixed combustion, the fuel and oxidizer are combined before ignition, forming a uniform mixture. This premixing can only occur at relatively low temperatures, where the chain-branching reactions responsible for sustaining combustion in the fuel are too slow to overcome the stabilizing effect of the chain-breaking reactions. Under these low-temperature conditions, the reaction mixture remains “frozen” or metastable. Combustion can then only start when an external heat source, such as a spark, raises the temperature high enough to trigger ignition. Once initiated, a flame front can propagate through the mixture as long as the equivalence ratio stays within the flammability limits [13]. The equivalence ratio ϕ is defined as

$$\phi = \frac{(F/O)_{\text{actual}}}{(F/O)_{\text{stoichiometric}}}, \quad (2.1)$$

where $(F/O)_{\text{actual}}$ and $(F/O)_{\text{stoichiometric}}$ denote the actual and stoichiometric fuel-oxidizer ratios, respectively. The fuel-oxidizer ratio is defined as the mass of fuel divided by the mass of oxidizer. Similarly, the fuel-air ratio represents the mass of fuel divided by the mass of air [12]. Also equivalently, the ratio of mass flow rates can be used [6]. The fuel-to-air ratio, and thus the equivalence ratio, can be directly adjusted with premixing to control how lean or rich the combustion is [13].

In nonpremixed combustion, the fuel and oxidizer are introduced into the combustion chamber separately and only mix as they burn through ongoing diffusion. This type of combustion is generally safer with respect to flashback risk and is therefore widely employed in applications such as furnaces, diesel engines, and gas turbines [13].

Between these two extremes lies partially premixed combustion, which frequently occurs in technical applications. In such cases, the fuel and oxidizer enter separately but undergo partial mixing due to turbulence or the geometry of the inlets. The practical advantage of this intermediate mode is that it combines beneficial aspects of both premixed and nonpremixed combustion while avoiding some of their adverse effects [13].

2.2 Turbulence Modeling

Since the combustion setup, which is considered in this work, takes place in a flowing medium, its behavior is described using the fundamental equations of fluid dynamics, namely the Navier-Stokes equations. Because most industrially relevant combustion flows are turbulent, direct resolution of all turbulent scales is infeasible, or extremely expensive. To address this, the flow variables are averaged. This averaging results in the Reynolds-Averaged Navier-Stokes (RANS) approach, where flow variables are decomposed into mean and fluctuating components [12].

Reynolds averaging is used for constant density flows. Since in chemically reacting flows density changes largely, for such flows, density-weighted (Favre) averaging is applied, where all the fluid mechanical quantities except the pressure are density weighted. The Favre averaging is defined as [12]:

$$\tilde{\phi} = \frac{\bar{\rho}\phi}{\bar{\rho}}, \quad (2.2)$$

with the decomposition of a variable ϕ

$$\phi = \tilde{\phi} + \phi'' . \quad (2.3)$$

Here, the overbar is the Reynolds average (as in $\bar{\rho}\phi$), the tilde the Favre average (as in $\tilde{\phi}$), and $''$ the Favre fluctuation (ϕ''). The Favre fluctuation satisfies [12]:

$$\bar{\rho}\phi'' = 0 . \quad (2.4)$$

Because of Equation 2.4, the Favre averaging of the governing mass, momentum and energy equations cancel out some of the terms. The resulting equations are [12]:

Mass conservation:

$$\frac{\partial \bar{\rho}}{\partial t} + \nabla \cdot (\bar{\rho} \tilde{\mathbf{u}}) = 0 . \quad (2.5)$$

Momentum conservation:

$$\frac{\partial(\bar{\rho} \tilde{\mathbf{u}})}{\partial t} + \nabla \cdot (\bar{\rho} \tilde{\mathbf{u}} \tilde{\mathbf{u}}) = -\nabla \bar{p} + \bar{\rho} \mathbf{g} + \nabla \cdot \bar{\boldsymbol{\tau}} - \nabla \cdot \widetilde{\bar{\rho} \mathbf{u}'' \mathbf{u}''} , \quad (2.6)$$

with

$$\bar{\boldsymbol{\tau}} = \mu(\nabla \tilde{\mathbf{u}} + (\nabla \tilde{\mathbf{u}})^\top - \frac{2}{3}(\nabla \cdot \tilde{\mathbf{u}}) \mathbf{I}), \quad (2.7)$$

and where $\widetilde{\bar{\rho} \mathbf{u}'' \mathbf{u}''}$ is the (density-weighted) Reynolds stress tensor.

Energy conservation:

$$\frac{\partial(\bar{\rho} \tilde{h}_t)}{\partial t} + \nabla \cdot (\bar{\rho} \tilde{\mathbf{u}} \tilde{h}_t) = \frac{\partial \bar{p}}{\partial t} + \nabla \cdot (\tilde{\mathbf{u}} \cdot \bar{\boldsymbol{\tau}}) - \nabla \cdot \bar{\mathbf{q}} - \nabla \cdot \left(\widetilde{\bar{\rho} \mathbf{u}'' h_t''} \right) , \quad (2.8)$$

where $\bar{\mathbf{q}}$ is the mean conductive/radiative heat flux, $h_t = h + \frac{1}{2}|\mathbf{u}|^2$ is the total specific enthalpy and $\widetilde{\bar{\rho} \mathbf{u}'' h_t''}$ is the turbulent heat flux.

Species transport (for species k):

$$\frac{\partial(\bar{\rho} \tilde{Y}_k)}{\partial t} + \nabla \cdot (\bar{\rho} \tilde{\mathbf{u}} \tilde{Y}_k) = -\nabla \cdot \bar{\mathbf{J}}_k - \nabla \cdot \left(\widetilde{\bar{\rho} \mathbf{u}'' Y_k''} \right) + \bar{\omega}_k, \quad (2.9)$$

where $\bar{\mathbf{J}}_k$ is the mean diffusive flux of species k , $\bar{\omega}_k$ is its mean production rate and $\widetilde{\bar{\rho} \mathbf{u}'' Y_k''}$ is the scalar flux.

This averaging process introduces additional unclosed terms, such as the (density-weighted) Reynolds stress tensor terms that represent the influence of turbulence on the mean flow, which cannot be determined directly and therefore require modeling. To account for these effects, turbulence closure models are employed, typically based on dimensional analysis and empirical correlations [13]. A common closure approach is the k - ϵ model, a two-equation model that solves transport equations for the turbulent kinetic energy k and its dissipation rate ϵ . This model efficiently represents the averaged influence of turbulence on the mean flow field and is widely applied in steady-state RANS simulations for combustion systems [12].

2.3 Nitrogen Oxides Formation

Nitrogen oxides (NO_x) primarily include nitric oxide (NO) and nitrogen dioxide (NO_2). These toxic and highly reactive gases are produced during the high-temperature ignition of fuels, originating either from the fixation of nitrogen in the combustion air or from nitrogen compounds contained within the fuel itself [14, 15].

Dry air consists mainly of nitrogen, oxygen, argon, carbon dioxide, and several other minor constituents known as trace gases [16]. Although molecular nitrogen (N_2) is generally inert and does not participate in chemical reactions under normal conditions, at temperatures above approximately 1600 °C it can dissociate. Through a sequence of reactions, this leads to the formation of nitric oxide (NO). This process is referred to as the thermal NO_x route or Zeldovich mechanism, named after Yakov Zeldovich (1945) [15]. The principal elementary reactions of the Zeldovich mechanism are:



and additionally in the case of water (H_2O) presence in the combustion environment



NO_2 is produced from the oxidation of NO in case of high temperatures. The rate of NO formation also increases strongly with temperature; thus, higher reaction temperatures promote greater production through the thermal route. However, this process is relatively slow compared to the rapid oxidative decomposition of hydrocarbon fuels. Consequently, long residence times at high temperatures favor increased NO_x formation. Therefore, strategies that reduce peak flame temperatures, limit oxygen availability or reduce the residence time of the hot combustion products are effective in curbing thermal NO_x generation [15].

In high-temperature regions, the thermal NO_x pathway represents the dominant mechanism of NO_x formation. However, other reaction routes can also contribute under certain conditions. These include the prompt NO route, which involves hydrocarbon radicals; the N_2O route, where nitrous oxide acts as an intermediate; the NNH route, involving nitrogen-hydrogen species; and the fuel-N route, where the nitrogen in fuel decomposes with the combustion process that produces NH_3 and HCN that would oxidate and produce NO [15].

2.4 Flashback Mechanisms

Flashback refers to the phenomenon in which the flame propagates from the combustion zone back into the premixing section of the combustor. This undesired propagation leads to a

significant increase in the temperature inside the premixer, which is not originally designed to withstand such thermal loads. As a result, flashback can cause severe damage to the engine and compromise operational safety. In gas turbine premixers, flashback can occur due to several mechanisms, of which four primary types are commonly distinguished [10].

The most relevant is boundary layer flashback (BLF), which occurs near the walls where the local flow velocity decreases due to the no-slip boundary condition. When the local burning velocity exceeds this reduced local flow velocity, the flame can propagate upstream along the wall. This mechanism is dominant in jet and Bunsen flames. Bunsen flames are simplified configurations frequently used to study laminar premixed flame propagation, produced by a Bunsen burner [10].

Non-uniform fuel-air mixing can strongly influence the occurrence of BLF, particularly near the walls. If the mixture is lean in the core but locally hydrogen-rich in the boundary layer, the equivalence ratio at the wall increases, leading to higher local flame speeds. Since the flow velocity near the wall is reduced by the no-slip condition, the local burning velocity may exceed it, allowing the flame to propagate upstream. This mechanism promotes BLF [6].

The second most relevant type is core flow flashback, which takes place when the turbulent burning velocity surpasses the local axial flow velocity in the core region of the combustor. Fuels with higher flame speeds, such as hydrogen, inherently increase the risk of this type of flashback compared to conventional hydrocarbon fuels [10].

Both of these flashback formation mechanisms rely on the imbalance of local flow velocity u and local flame (burning) velocity $S_f(y)$. In a premixed flame, the flame front propagates into the unburned mixture at the local flame velocity. At the same time, the incoming flow moves in the opposite direction, carrying the unburned mixture toward the flame with a local flow velocity. Under stable operation, these two effects balance: The flow velocity pushes the unburned mixture toward the flame, and the flame propagates against the flow at the same rate. Figure 2.1 [17] demonstrates this phenomenon. However, when the local burning velocity exceeds the local flow velocity, the flame front can advance upstream faster than the fresh mixture is being supplied, resulting in flashback [17].

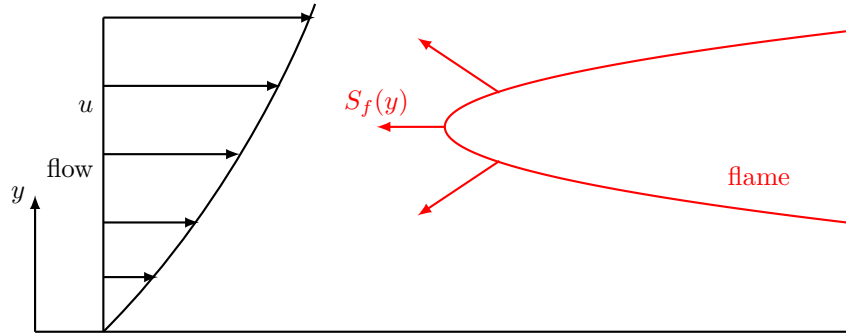


Figure 2.1: BLF

Other mechanisms include combustion-instability-induced flashback, which can result from large-amplitude fluctuations of the flow field. Such instabilities may arise from the interaction of acoustic modes, unsteady heat release, or flow structure variations. Pressure and velocity oscillations can cause flame movement, generate large vortices, and, in severe cases, induce flow reversal leading to flashback [10].

Finally, combustion-induced vortex breakdown (CIVB) may occur in swirl-stabilized gas tur-

bine combustors. When the swirl number, representing the relative strength of the swirl flow, exceeds a critical value, a vortex breakdown can form with a stagnation point and recirculation bubble. If this bubble moves upstream, it can trigger flashback into the premixer [10].

Several flashback mitigation strategies have been proposed for diffusion and premixed burners in the literature. For instance, micromixing combustion technology employs numerous micronozzles with very small geometrical dimensions to generate many tiny, spatially separated flames. This configuration helps maintain lower local flame temperatures and shortens the residence time of hot combustion gases, thereby reducing NO_x formation and simultaneously the likelihood of flashback, since the flame front cannot easily propagate back into the fuel inlets. Nevertheless, while this approach is effective in limiting core flow flashback and combustion-induced vortex breakdown, it does not eliminate the danger of boundary layer flashback, where near-wall flame propagation can surpass the low inlet velocity and initiate flashback [6, 17].

2.5 The NO_x -Flashback Tradeoff in Hydrogen Combustion

Hydrogen combustion is characterized by exceptionally high flame temperatures [18, 19]. Because hydrogen contains neither nitrogen nor carbon in its molecular structure, the formation of nitrogen oxides occurs mainly through the thermal NO_x pathway, following the Zeldovich mechanism [15]. The reaction temperature plays a decisive role: under identical stoichiometric conditions, the adiabatic flame temperature of a hydrogen–air mixture (2207 °C) is considerably higher than that of methane (1917 °C) [19].

To mitigate NO_x generation, lean premixed combustion is often applied. When hydrogen is burned under lean conditions, the flame temperature decreases significantly because the mixture contains more air and fewer fuel-rich zones. Moreover, hydrogen’s low flammability limit allows for stable combustion even at very high air dilution levels. As a result, NO_x formation can be drastically reduced [19].

Hydrogen combustion also introduces a higher risk of flashback [18]. Due to its high laminar flame speed, the hydrogen flame can propagate upstream more easily than that of methane [19]. Another phenomenon contributing to hydrogen’s instability is the occurrence of turbulent diffusion instabilities (TDIs). These arise from the low Lewis number of hydrogen, which reflects its high molecular diffusivity compared to the thermal diffusivity of the mixture. This enhances small flame front perturbations, leading to strongly wrinkled flame surfaces and increased fuel conversion rates. The high diffusivity of hydrogen also leads to locally richer or leaner regions that can promote early ignition [20]. Since hydrogen–air mixtures also ignite very easily, even small hot spots or recirculation zones can trigger upstream flame propagation. Also, because hydrogen burns over a wide equivalence ratio range, it remains combustible even in lean or inhomogeneous mixture regions where other fuels would be too lean to ignite [19].

The overall degree of premixing also influences flashback behavior. In completely premixed systems, the flame is no longer confined to a diffusion zone and can propagate wherever the mixture is flammable. This effect is intensified for hydrogen because of its high diffusivity and reactivity. This means; while a uniform hydrogen distribution is beneficial for reducing NO_x emissions, it simultaneously increases flashback susceptibility. Conversely, nonpremixed systems are inherently more stable and safer, since fuel and oxidizer meet only at the flame front, preventing upstream ignition. However, as mentioned before, these configurations tend to produce higher NO_x emissions [21, 22].

In essence, lean premixed hydrogen combustion presents a fundamental trade-off: it achieves low NO_x emissions but exhibits a high flashback risk. These are conflicting design objectives, requiring a systematic and quantitative approach to identify burner geometries that achieve an

optimal compromise between stability and emission performance.

2.6 Design Optimization

For the optimization of a function over a continuous design space, three primary approaches are commonly discussed in the literature: gradient-based, gradient-free, and surrogate-based optimization methods. In gradient-based methods, calculating the gradient poses a significant challenge for CFD-based optimization. Although adjoint solvers can provide gradients at a relatively lower computational cost, their implementation becomes highly complex in reactive flow simulations, where non-differentiable source terms arise from chemical reactions. On the other hand, gradient-free approaches, such as genetic and evolutionary algorithms, typically require an excessive number of CFD evaluations to achieve convergence, making them computationally expensive [23].

Surrogate-based optimization addresses the high cost of CFD evaluations by replacing or augmenting the expensive model with a cheaper approximation, called “surrogate” that mimics its behavior. The optimization is then driven on the surrogate, with occasional sampling (i.e. CFD or experimental evaluations) to correct or refine it [24].

One very powerful surrogate-based framework is Bayesian Optimization, where the core idea is to build a probabilistic surrogate model of the objective function, and then maximize an acquisition function to decide where to sample next [25].

The probabilistic surrogate model is commonly chosen as a Gaussian Process (GP), which is a probabilistic nonparametric regression model that defines a prior over functions. Given observed data x_i, y_i , it produces a posterior predictive distribution $N(\mu(x), \sigma^2(x))$ at any new input x . Here, $\mu(x)$ is the predicted mean, and σ^2 is the predictive standard deviation that quantifies the uncertainty. The utility of a GP in Bayesian optimization comes from the fact that it naturally quantifies uncertainty, enabling principled trade-offs between exploration, e.g. sampling where uncertainty is large and exploitation, e.g. sampling where the predicted objective is good, when the acquisition function is used for sampling [25].

The acquisition function $\alpha(x)$ quantifies how “useful” sampling at x would be, by balancing exploration and exploitation. A widely used acquisition function is Expected Improvement (EI). EI estimates, in expectation, how much better the objective at x could be relative to the best current solution, accounting for both the predicted mean and uncertainty from the surrogate. In other words,

$$\alpha_{\text{EI}}(x) = \mathbb{E}[\max(0, f_{\text{best}} - f(x))], \quad (2.13)$$

where f is the true (unknown) objective function to be minimized and f_{best} is the best observed value so far. Since the surrogate model is chosen as a GP, the true (unknown) objective f can be assumed to be normally distributed. Thus, the EI can be computed as:

$$\alpha_{\text{EI}}(x) = \begin{cases} (f_{\min} - \mu(x)) \Phi(Z) + \sigma(x) \phi(Z), & \text{if } \sigma(x) > 0, \\ 0, & \text{if } \sigma(x) = 0, \end{cases} \quad (2.14)$$

where

$$Z = \frac{f_{\min} - \mu(x)}{\sigma(x)}. \quad (2.15)$$

Here, $\mu(x)$ is the predicted mean of the surrogate model at point x , $\sigma(x)$ is the predicted standard deviation of the surrogate model at point x , $\Phi(Z)$ is the cumulative distribution

function (CDF) of the standard normal distribution, and $\phi(Z)$ is the probability density function (PDF) of the standard normal distribution [25].

Therefore, regions with high uncertainty ($\sigma(x)$) or good predicted performance/low predicted mean ($f_{\text{best}} - f(x)$) tend to yield high EI. In the initial iterations, new points are added in parameter regions that were previously less explored, representing an exploratory stage aimed at mapping the overall design space. Later, the concentration of samples in areas that offer the greatest improvements indicates a shift toward exploiting these promising regions [6, 25].

The Bayesian optimization procedure iterates as follows [25]:

1. Start with an initial sampling of design points (via e.g. Latin Hypercube or space-filling design).
2. Fit a surrogate GP to the observed input–output data.
3. Based on the surrogate’s predictive mean and predictive uncertainty, evaluate an acquisition function $\alpha(x)$ over the design space.
4. Choose the next sample point $x_{\text{new}} = \arg \max \alpha(x)$ and evaluate the true expensive objective $f(x_{\text{new}})$.
5. Augment the dataset with $(x_{\text{new}}, f(x_{\text{new}}))$ and refit/update the surrogate.
6. Repeat steps 3–5 until a termination criterion is met (budget, convergence, etc.).

Bayesian optimization is particularly attractive for CFD-based design tasks, where each function evaluation is a simulation run that can be computationally expensive. In such optimization tasks, a parametric CAD model or geometry parameterization is first defined, in which key geometric quantities such as inlet dimensions, angles, or curvature parameters serve as design variables. These parameters constitute the input vector x to the black-box objective function, whose evaluation involves a CFD-based combustion simulation that provides the resulting performance metrics (e.g., NO_x emissions, pressure loss, or flashback indicators). Because surrogate models can approximate the system response using a limited number of high-fidelity evaluations, the total computational cost of the design process is substantially reduced compared to direct or brute-force optimization methods [6].

In a multi-objective setting, more than one objective function $f_1(x), f_2(x), \dots$ is used. Bayesian multi-objective optimization uses surrogates for each objective (or a joint surrogate) and acquisition strategies designed for trade-offs, e.g. Pareto front exploration, ensuring sampling focuses on regions likely to improve the Pareto front, also called Pareto frontier, which represents the set of solutions that are non-dominated, meaning that no other design performs better in all objectives simultaneously. Each point on the front expresses an optimal trade-off between conflicting goals, for example minimizing cost while maximizing efficiency, where improving one inevitably worsens another. Instead of a single optimum, multi-objective Bayesian optimization aims to efficiently approximate this frontier [25]. Figure 2.2 illustrates the principle of the Pareto front.

An example for gradient-free optimization methods are genetic algorithms (GAs), inspired by principles of biological evolution. Unlike gradient-based optimization strategies that require derivative information, or surrogate-based methods that need large amount of sampling data at start to build the surrogate model, GAs rely solely on function evaluations and stochastic exploration mechanisms. This makes them suitable for problems that are noisy, non-smooth, discontinuous, or highly nonlinear. The terminology used in GAs, such as population, generation, fitness, mutation, and crossover, originates from genetics and evolutionary biology [26, 27].

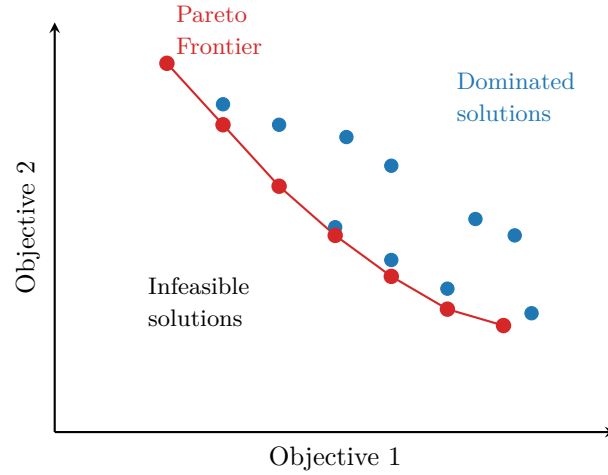


Figure 2.2: Illustration of the Pareto frontier

In a GA, a population represents a set of candidate design points in the design space. Each iteration of the algorithm is referred to as a generation, and a subpopulation may be formed for specific evolutionary operations. A design point is said to survive into the next generation if it is selected based on its fitness, i.e., its quality with respect to the optimization objectives. Groups of points that cluster in similar regions of the design or objective space are called niches. These help maintain population diversity and prevent premature convergence [26].

Genetic algorithms can be extended naturally to multi-objective optimization, resulting in multi-objective genetic algorithms (MOGAs). In multi-objective problems, the goal is not to find a single optimal design but, comparable to multi-objective surrogate-modeling approaches, to approximate and converge to the Pareto-optimal set of solutions [26, 27].

Evolutionary algorithms are particularly suitable for multi-objective optimization because they maintain a set of solutions at each iteration. This allows the algorithm to identify several Pareto-optimal candidates within a single run, eliminating the need for repeated single-objective optimizations. Moreover, MOGAs can efficiently handle non-convex, discontinuous, or even disconnected Pareto fronts, which often pose difficulties for gradient-based or surrogate-based optimization approaches [27]. Also, there is no need for a multi-dimensional surrogate model to be built, which makes multi-objective surrogate modeling complex to implement.

However, this flexibility comes at a cost. MOGAs typically require a large number of function evaluations [27]. When each evaluation corresponds to a high-fidelity simulation, such as a RANS inlet flow simulation combined with a reacting-flow outlet simulation, the computational expense can become significant.

Developing a MOGA involves addressing three key questions [26]:

1. How should the fitness of each design point be evaluated?
2. How should Pareto optimality be incorporated into the selection mechanism?
3. Which design points should survive into the next generation?

In multi-objective GAs, fitness is typically determined using dominance concepts, crowding measures, or ranking mechanisms. These methods aim to promote nondominated solutions while preserving population diversity [26]. MOGA offers the flexibility to not be dependent on a-priori information and therefore may begin with a randomly generated or Latin Hypercube-sampled

initial population and iteratively improves the population until it approaches the trade-off Pareto surface between objectives [27].

Considering the discussed points about MOGA, the general procedure looks as follows:

1. Begin by generating a set of random design candidates based on the defined variable bounds. Each candidate corresponds to one possible inlet geometry.
2. For every candidate, evaluate the objective.
3. Determine which designs dominate others. Non-dominated designs form the current approximation of the Pareto front. A fitness value is assigned based on this dominance ranking.
4. Select parent designs: Designs with better fitness (i.e., closer to the Pareto front) are more likely to be selected as parents for generating the next generation.
5. Create new designs via crossover and mutation: Generate offspring by mixing parent variables (crossover) and randomly altering some variables (mutation). This produces new inlet geometries that have not been evaluated before.
6. Evaluate offspring, in which each new design triggers another full simulation cycle. These evaluations dominate the overall computational cost.
7. Combine parents and offspring, then select the best and most diverse set of designs to form the new population.
8. Repeat steps 2-7 until a termination criterion is met (budget, convergence, etc.).

Several well-established MOGA variants are commonly used in engineering optimization, like NSGA-II (Non-dominated Sorting Genetic Algorithm II) that uses nondominated sorting and crowding distance and is an improvement from NSGA [28], SPEA2 (Strength Pareto Evolutionary Algorithm 2) that incorporates an external archive of nondominated solutions [29], and PAES (Pareto Archived Evolution Strategy) that is a simple mutation-based strategy with an external archive [30].

Additive manufacturing (AM) fits well with surrogate-based, as well as genetic-algorithm-based automated optimization loops, because AM allows layer-wise fabrication of complex internal features without the constraints of conventional tooling. The design freedom of AM means that the optimized geometries from an optimization loop can often be manufactured directly, without the design-for-manufacturability compromises typical of subtractive or conventional methods [6]. AM is particularly beneficial for the development and optimization of NO_x reduction strategies, such as tangentially arranged fuel injectors that introduce a mild swirl. The induced swirl breaks the flame into smaller structures and shortens the residence time of hot gases, thereby decreasing NO_x emissions as the swirl intensity increases. Similarly, micromixing burners, as previously mentioned, also help minimize NO_x formation and flashback risk. While these approaches show promise for achieving safe and efficient hydrogen combustion, they often involve intricate geometries that are challenging to produce using traditional manufacturing techniques such as milling or casting [6].

Chapter 3

State of the Art

3.1 The Baseline Optimization Framework

The developed framework in this thesis is based on the work by Juris et. al. [6]. This study introduces an automated design optimization framework developed for hydrogen burner geometries, specifically focusing on the inlet region. According to the authors, the inlet geometry plays a decisive role in flame stabilization and temperature distribution, which govern both the risk of flame flashback and the level of NO_x emissions. As discussed in Chapter 2, improved mixing quality leads to a more homogeneous equivalence ratio field, and thus lowers thermal NO_x formation. Authors also argue that enhanced mixing prevents the formation of a lean core surrounded by locally hydrogen-rich boundary layers, which would otherwise cause locally high equivalence ratio values near the walls. Therefore, improved mixing also consequently increases the likelihood of flashback in the specific example of cross-flow hydrogen combustion. For these reasons, the overall objective of the framework is to develop an additively manufacturable, partially premixed cross-flow jet burner for hydrogen combustion that achieves superior mixing and, as a result, ensures both flame stability and low NO_x emissions [6].

To achieve this objective, the framework establishes a fully automated optimization loop coupling Dakota, Salome, and OpenFOAM; three open-source tools that sequentially enable data-driven optimization, geometry parametrization, and CFD evaluation. The overall workflow, as shown in Figure 3.1, is coordinated by Dakota, which manages the generation of design parameters, links geometry creation in Salome with numerical simulations in OpenFOAM, and automates the iterative evaluation of candidate designs. Within this loop, Salome is employed for parametric CAD modeling. It supports both graphical and script-based modeling through its Python interface, and allows automatic geometry generation and meshing based on the input design parameters. The resulting meshes are subsequently used in OpenFOAM, which is a modular CFD toolbox capable of simulating reactive flow, heat transfer, and mixing phenomena. Simulation results are then passed back to Dakota, which updates the design space exploration and guides subsequent iterations [6].

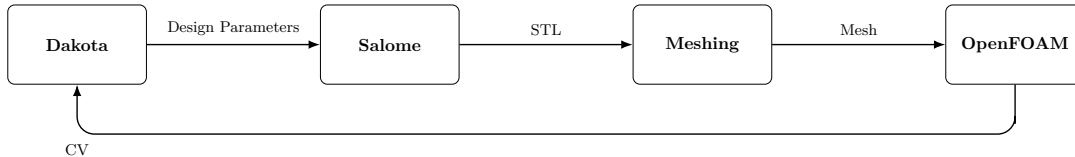


Figure 3.1: Optimization Framework Architecture

The design optimization relies on the Efficient Global Optimization (EGO) algorithm implemented in Dakota. EGO is a Bayesian optimization approach that constructs a GP surrogate

model to approximate the expensive CFD-based objective function. This method enables targeted refinement of the surrogate in promising regions of the design space while requiring comparatively few CFD evaluations. For the initial sampling of the parameter space, the framework employs Latin Hypercube Sampling (LHS), a stratified sampling technique that ensures a uniform and space-filling distribution of sample points compared to purely random sampling. Following the recommendations of the Dakota documentation, the number of initial samples typically ranges between two and ten times the number of design parameters, defining the “wind-up phase” required for the surrogate model to reach an adequate level of representativeness [6].

After the initial sampling, the EGO optimization proceeds through a sequence of user-defined iterations in which new design candidates are proposed based on the Expected Improvement Function serving as the acquisition criterion. In each iteration, the design configuration predicted to yield the highest expected improvement is evaluated through a new CFD simulation. The resulting data are then incorporated into the Gaussian Process surrogate model, continuously refining its predictive accuracy in regions of high potential. This adaptive sampling strategy ensures efficient convergence toward the optimum while keeping the computational cost manageable [6].

3.2 Application Example of the Baseline Framework

To demonstrate the functionality and robustness of the proposed optimization framework, the reference study [6] applies it to a simplified burner configuration inspired by a jet-in-crossflow (JICF) system. In this setup, multiple fuel jets are injected transversely into a main oxidizer stream within a confined duct. The design variables considered in the optimization are the fuel inlet angle α , the inlet height h , the inlet diameter d , and the number of fuel inlets $N_{\text{fuel_inlets}}$.

The objective function used in the optimization is the coefficient of variation (CV) of the equivalence ratio, computed over all outlet cells, which quantitatively describes the degree of mixing:

$$\text{CV} = \frac{\sigma_\phi}{\mu_\phi}, \quad (3.1)$$

where σ_ϕ and μ_ϕ denote the standard deviation and mean of the local equivalence ratio ϕ , respectively. A lower CV corresponds to a more homogeneous mixture and thus indicates improved mixing performance. As a dimensionless and statistically robust metric, CV captures both the magnitude and spatial consistency of mixture variations and is therefore well suited for comparing flow configurations with differing absolute equivalence ratios [6].

Each of the optimization parameters has a direct influence on the resulting flow field and mixing behavior. For instance, the inlet height h controls the available mixing length: larger heights allow more time for the fuel and oxidizer streams to homogenize before exiting the duct, thereby improving mixture uniformity and potentially reducing NO_x formation [6].

For numerical stability and to ensure consistent boundary conditions across design variations, the geometries generated in Salome include extended inlet and outlet sections. The inlet section is prolonged by 5 mm to allow the velocity profile to fully develop before entering the mixing region, while the outlet section is extended by 15 mm to suppress artificial recirculation and minimize backflow at the boundary. These extensions improve the robustness of the numerical solution by damping transient effects and ensuring that the monitored flow field within the mixing region remains physically consistent across different geometries [6].

A parametric CAD model is constructed in Salome to automatically reflect variations in

the four design parameters. This model serves as the geometric foundation for the automated optimization process. The chosen parameters are selected based on their strong physical influence on jet penetration, recirculation zone formation, and global mixing efficiency, which are critical for achieving a homogeneous fuel-air mixture prior to combustion [6].

The CFD simulations are conducted using OpenFOAM, employing the steady-state Reynolds-Averaged Navier–Stokes (RANS) equations and the $k-\epsilon$ turbulence model with the `reactingFoam` solver. Although `reactingFoam` is designed for compressible, turbulent, reacting flows, in this study the combustion model is deactivated to isolate and investigate the non-reacting hydrogen–air mixing process. The simulations are performed at a global Reynolds number of 7000 and an equivalence ratio of 0.6. The scalar transport is modeled using the `unityLewisFourier` model, which assumes equal Lewis numbers for all species, a significant but common simplification in hydrogen-air mixing studies [6].

Boundary conditions, as described in the Table 3.1, are prescribed consistently across all simulation cases to ensure comparability. Here, $\dot{Q}$ indicates the volume flow. The computational mesh is generated with a base cell size of 0.14 mm and an expansion ratio of 1.2, providing adequate spatial resolution of the main shear layers while maintaining manageable computational cost. To capture near-wall gradients, boundary layer refinement is applied: five prism layers with a 55% growth ratio are used for the oxidizer channel walls, while the smaller and more curved fuel inlets are refined with three layers at a 30% growth ratio. All mesh parameters are kept constant throughout the optimization, as manual retuning is incompatible with the automated workflow [6].

To assess the surrogate model’s predictive capability, the framework applies a Leave-One-Out (LOO) cross-validation scheme using the Mean Absolute Percentage Error (MAPE) as the evaluation metric:

$$\text{MAPE} = \frac{100\%}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right|, \quad (3.2)$$

where y_i and $\hat{y}_i$ denote the CFD-derived and surrogate-predicted objective values, respectively. MAPE is chosen because it is scale-independent and robust for low-magnitude targets, enabling a fair comparison across varying output scales. Finally, among all evaluated designs, the configuration with the lowest CV value from the CFD simulations is selected as the optimal geometry [6].

Table 3.1: Boundary conditions for oxidizer and fuel inlets and outlet [6]

	Inlet Oxidizer	Inlet Fuel	Outlet
U	122 m/s	calculated	zero gradient
ρ	1.2 kg/m ³	0.084 kg/m ³	-
$\dot{Q}$	6.0367×10^{-4} m ³ /s	1.5217×10^{-4} m ³ /s	-
T	298 K	298 K	-
p	zero gradient	zero gradient	total pressure

For time integration, a local Euler scheme with dynamically adjustable timesteps is employed, adapting to local Courant number conditions to maintain numerical stability. The initial design space exploration is performed using Latin Hypercube Sampling with ten samples per design variable, resulting in a total of 40 CFD simulations that form the initial surrogate model [6].

Finally, upper and lower bounds for each design variable are defined based on geometric feasibility, additive manufacturability, and numerical stability constraints, as shown in Table 3.2.

These limits ensure that the optimized designs remain physically meaningful and producible within the intended application context [6].

Table 3.2: Design variable limits for the jet-in-crossflow burner configuration [6]

	α [°]	h [mm]	d [mm]	$N_{\text{fuel_inlets}}$
Upper limit	40	10	1.1	3
Lower limit	70	20	1.6	6

The results of the optimization framework, used on the presented application example, are shown in Table 3.3. The optimized configuration yields a coefficient of variation CV that is 52.2% lower than the mean CV obtained across all evaluated designs, demonstrating a significant improvement in mixing uniformity. The surrogate model initially exhibits a MAPE of 19.86% based on the 40 LHS evaluations, which progressively decreases to 16.76% after an additional 32 EGO iterations. This trend reflects a continuous enhancement in the model’s predictive capability for estimating CV. Furthermore, the authors report that the automated optimization framework achieves an approximate 87.5% reduction in computational cost compared to a full-factorial design of experiments with identical parameter bounds, which highlights its efficiency in exploring complex design spaces [6].

Table 3.3: Parameter sets of optimal and mean CV [6]

	α [°]	h [mm]	d [mm]	$N_{\text{fuel_inlets}}$	CV
Optimal Design	40	10	1.53	6	0.0098
Design with mean CV	65	18.3	1.29	4	0.0204

3.3 Unresolved Aspects of the Baseline Framework

While the proposed framework establishes a systematic and fully automated approach to geometry optimization for hydrogen burner configurations, the authors acknowledge several restrictions and unresolved aspects that limit its predictive accuracy and general applicability.

First, all key performance metrics in the baseline study are derived from temporally averaged RANS data. Although this approach is computationally efficient, it inherently suppresses localized and transient fluctuations in the equivalence ratio field. Such fluctuations, however, are critical for predicting flashback initiation and hot-spot-driven NO_x formation. Consequently, the framework does not capture the time-dependent dynamics of the mixing process, which could strongly influence the burner’s real-world behavior under unsteady operating conditions.

Furthermore, the authors note that the optimal solution obtained in their study lies at the boundaries of the defined design space, specifically, at the lower limits of the fuel inlet angle α and height h , and at the upper limits of the inlet diameter d and number of inlets $N_{\text{fuel_inlets}}$. This boundary convergence indicates that potentially more favorable configurations may exist beyond the explored parameter ranges, suggesting that the current design space definition may not yet encompass the true global optimum.

Another limitation arises from the use of a surrogate-based model trained on a limited sample budget. Although Bayesian optimization minimizes the number of high-fidelity CFD evaluations, the resulting surrogate model accuracy is constrained by the small number of training samples,

which can introduce prediction errors or local overfitting.

Moreover, the absence of reactive CFD simulations means that combustion-related effects, particularly localized equivalence-ratio peaks that act as indicators of flashback risk, are not captured in the optimization loop.

The framework also lacks experimental validation. No physical testing or measurement of the optimized geometries fabricated via additive manufacturing has been conducted. Such validation would be essential to quantify the deviation between simulated and real-world performance, as CFD results are always limited by turbulence modeling assumptions, discretization errors, and idealized boundary conditions.

Lastly, the optimization targets are defined only in terms of mixing homogeneity, with flashback risk and NO_x formation not explicitly included as objective functions. This indirect optimization approach assumes that improved mixing inherently reduces NO_x and flashback tendencies, which may not always hold true in practical reacting systems.

Building upon these limitations, the present thesis aims to expand the original framework in several key directions. The extended methodology will include reactive CFD simulations, enabling the explicit capture of combustion-related phenomena and local equivalence-ratio peaks. Additionally, NO_x detection mechanisms will be incorporated, providing a direct quantitative measure of pollutant formation within the flow field. In parallel, a flashback detection criterion will be introduced, allowing the formulation of an explicit objective function for assessing flashback risk. Finally, the optimization process itself will be advanced toward a multi-objective optimization framework, where NO_x emissions and flashback risk serve as primary objectives, rather than relying solely on mixing homogeneity as a proxy.

3.4 Improvement Strategies for the Optimization Framework

3.4.1 Flashback Detection

The central unresolved aspect of the baseline framework [6] is the absence of a direct flashback detection mechanism embedded within the CFD solver. Such a capability would enable the solver to automatically transfer flashback-related information to the optimization framework. In particular, boundary layer flashback (BLF) remains especially relevant to detect and quantify, since it cannot be effectively mitigated by conventional stabilization methods such as micromixing, as stated in Chapter 2.

In the literature, various criteria have been formulated to predict the onset of BLF. The first well-established theoretical approach to predict BLF is the Critical Velocity Gradient (CVG) criterion, originally introduced by Lewis and von Elbe [17, 31]. The concept suggests that flashback occurs when the flow velocity gradient near the wall becomes smaller than the velocity gradient associated with flame propagation, typically at the so-called penetration distance δ_p . The flow velocity close to the wall is assumed to vary linearly, and thus the local velocity gradient can be expressed as:

$$g = \left. \frac{du}{dy} \right|_{y=y_{\text{wall}}} . \quad (3.3)$$

At the penetration distance δ_p , the local flow velocity equals the characteristic laminar flame speed S_f , defining the critical gradient as:

$$g_c = \frac{S_f}{\delta_p} . \quad (3.4)$$

Laminar flame speed S_f , penetration distance δ_p and the local velocity gradient are demonstrated in Figure 3.2 [17].

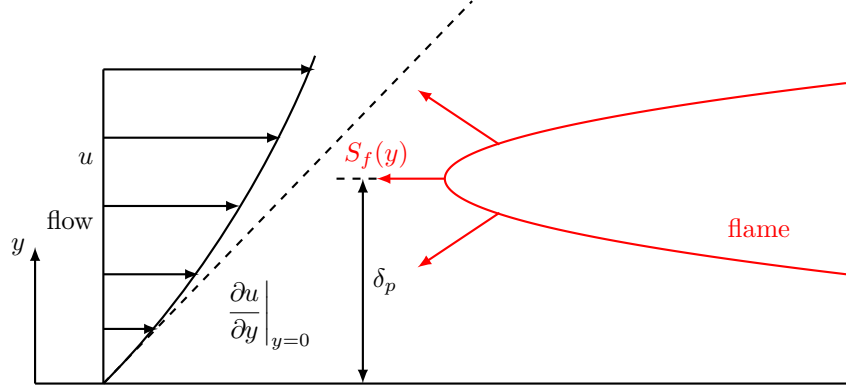


Figure 3.2: BLF and CVG

If the ratio of these two gradients exceeds unity,

$$\frac{g_c}{g} > 1, \quad (3.5)$$

a flashback is predicted to occur. In this context, S_f and δ_p must be modeled from the local flow and thermochemical conditions [17]. However, this approach neglects detailed flame–flow interactions and assumes a laminar velocity distribution near the wall, limiting its applicability to laminar pipe flows [17, 32].

Lin et al. [17, 32] extends the original CVG approach to turbulent conditions by introducing a turbulent-flame-speed-based critical velocity gradient (T-CVG). In their formulation, the characteristic velocity S_f is replaced by the turbulent flame speed S_T evaluated at the penetration distance. Furthermore, the penetration distance δ_p is assumed to scale with the laminar flame thickness under inlet conditions. To incorporate differential diffusion effects, a Lewis number term Le_{fuel} is introduced, defined as the ratio of thermal diffusivity to the mass diffusivity of the deficient fuel species:

$$Le_{\text{fuel}} = \frac{\alpha}{D_{\text{fuel}}}. \quad (3.6)$$

The resulting critical velocity is expressed as:

$$g_c = \frac{S_T}{Le_{\text{fuel}} \delta_p}. \quad (3.7)$$

Flashback occurs when $g_c/g > 1$, analogous to the laminar CVG criterion. Although this model accounts for turbulence through S_T , it neglects the influence of flame stretch, which can dominate near-wall conditions in many BLF scenarios where turbulence intensity is relatively low.

To better represent the physical processes responsible for boundary layer flashback, Novoselov et al. [17, 33] proposed the Combined Stretch and Heat-Loss (CSF) criterion. This approach incorporates the coupled effects of flame stretch and thermal losses, yielding a more general condition for the onset of BLF:

$$g_c = f(S_{L,\text{ext}}, \alpha, g). \quad (3.8)$$

Here, $S_{L,\text{ext}}$ denotes the laminar flame speed at extinction conditions, and α represents the thermal diffusivity of the mixture. The CSF criterion can be interpreted as a competition

between two characteristic time scales: the near-wall flow time scale ($1/g$) and the thermal flame time scale ($\alpha/S_{L,ext}^2$). When the thermal flame time scale becomes shorter than the flow time scale, i.e. $g_c/g > 1$, the flame front can penetrate the boundary layer and flashback occurs. This formulation provides improved physical fidelity compared to the CVG family of criteria by explicitly considering both stretch and quenching mechanisms.

Building upon both the T-CVG and CSF formulations, Zhang et al. [17] developed the Extended Turbulent CVG (ET-CVG) criterion, which integrates the effects of turbulence, stretch, and heat loss in a unified framework. Two main enhancements distinguish this model from its predecessors: First, a turbulent flame scaling law is introduced to represent near-wall stretch effects on the flame front. Second, the Lewis number of the deficient fuel species in the T-CVG model is replaced by the effective Lewis number Le_{eff} , which accounts for the combined diffusive behavior of all participating species and thus provides a more accurate description of differential diffusion in lean hydrogen–air mixtures.

The ET-CVG criterion is generally written as:

$$g_c = \frac{S_T}{Le_{eff} \delta_p} f_{stretch}, \quad (3.9)$$

where $f_{stretch}$ is a scaling factor derived from the flame-stretch correlation. As with the previous criteria, a local flashback risk is identified, when $g_c/g > 1$. By combining the turbulence, diffusion, and stretch effects, the ET-CVG formulation provides a more comprehensive framework for flashback prediction, suitable for implementation in advanced CFD-based optimization workflows.

3.4.2 NO_x Detection

A second unresolved aspect for closing optimization loop is the in-place prediction of nitric oxide emissions inside the CFD solver, so that the optimizer can directly consume a scalar NO_x metric. Because in lean premixed hydrogen flames, the dominant pathway for thermal NO_x formation is the Zeldovich mechanism [10], implementing this pathway within `reactingFoam` enables solving an additional transport equation for the NO_x mass fraction and to post-process an objective in design optimization. An example of an in-place NO_x implementation and analysis in a pent-roof spark-ignition engine can be found in the work by Sencic et al. [34].

Chapter 4

Methodology

Building upon the combustion modeling principles and the optimization strategy discussed in the preceding chapters, this work adopts a methodological framework that remains closely aligned with the baseline optimization case while extending it to reactive conditions. The same set of design parameters, as well as identical boundary and initial conditions defined in the baseline framework are retained. Likewise, the geometric configuration of a simplified 3D burner derived from a jet-in-crossflow arrangement used in the reference study is kept unchanged. This geometry consists of multiple transverse fuel inlets discharging into a main oxidizer stream within a confined duct, thereby capturing the dominant mixing mechanisms of practical hydrogen burners without introducing the full complexity of industrial burner assemblies [6]. Due to this configuration, the probability of partially premixed regions is inherently high.

However, because the objective of this thesis is to evaluate flashback risk and NO_x formation under reactive operation, the non-reacting baseline workflow is extended by introducing a second simulation stage. So the first stage, or the inlet simulation, mirrors the non-reacting setup of the baseline paper to resolve the mixing field upstream of the flame, while the second stage, or the outlet simulation, is implemented as a 2D reactive domain into which the outlet profile of the inlet simulation is mapped. In this domain, the `multicomponentFluid` solver and a detailed hydrogen oxidation mechanism are used to capture the NO_x chemistry and the flame stabilization, thereby enabling reactive CFD analysis in the workflow. The flame enters this 2D domain through `inlet1`, which serves as the interface between the two simulation stages, as shown in Figure 4.1.

Together, this two-stage approach allows to examine the thermochemical phenomena in the outlet of the baseline hydrogen burner without having to change the original simulation setup, while also providing a modular setup that enables the analysis of other properties in the outlet for further research.

4.1 Outlet Simulation

The outlet simulation is conducted using steady-state RANS with pseudo-time stepping and the standard $k-\epsilon$ turbulence model, implemented through the `multicomponentFluid` solver in OpenFOAM, which extends perfect-gas mixtures to multicomponent reactive systems and allows all species of the combustion mechanism to be included consistently. This model enables the computation of diffusion, enthalpy and production terms for all species in the hydrogen-air- NO_x mechanism. The global Reynolds number is chosen as $\text{Re} = 10000$.

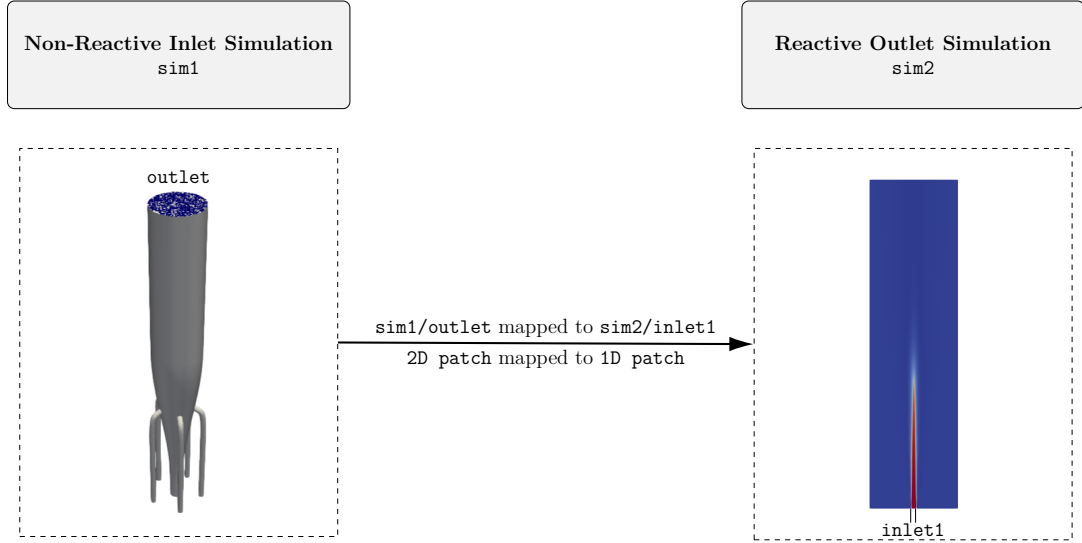


Figure 4.1: The two simulation stages

The `multicomponentFluid` solver is configured with the Eddy Dissipation Concept (EDC) combustion model. The EDC framework links turbulence and chemistry by assuming that reactions occur inside fine-structure regions determined by the local turbulence time scales [35].

The transport properties are evaluated using Sutherland’s law. The Sutherland constant A_s is set to $1.512 \times 10^{-6} \text{ kg/ms} \sqrt{\text{K}}$, and the Sutherland temperature T_s is taken as 120 K. Furthermore, NASA JANAF thermodynamic polynomials are used to represent the temperature-dependent specific heats, enthalpies, and entropies of the species. The equation of state is set to the perfect-gas law. The fluid is assumed to be Newtonian. In all simulations, a unity Lewis number is assumed, meaning that mass and thermal diffusivities are treated as equal for all species. Since the Lewis number of hydrogen is approximately 0.3 in mixtures with air [36], this is a strong simplification. Nevertheless, this assumption is made to stabilise the numerical solution and avoid excessive stiffness. The chosen turbulent Prandtl number is 0.85.

The overall boundary of the outlet domain is divided into eight distinct patches, corresponding to the geometry previously shown in Figure 4.1. The flame enters the 2D domain from `inlet1`. `inlet2` provides an additional co-flow. The extruded faces of the 2D plane are represented by `frontAndBack`, which stays empty. The boundary patch `wall`, depicted in Figure 4.2, is introduced at the inlet to render the geometry suitable for flashback assessment. Two slits, between which lies the boundary patch `inlet1`, create two walls in the flow inlet. These walls allow the flashback risk to be quantified by evaluating and comparing the velocity gradients along the wall surfaces, as described in Chapter 2. The artificial upper and lower walls are described as `upperWall` and `lowerWall`. There is one `outlet`, where the outflow conditions are applied. A summary of the applied boundary conditions for the patches is given in Table 4.1, where “default” refers to all the species that are not explicitly listed in the table. The initial conditions (IC) for the internal field are also given in Table 4.1.

For k , a turbulent intensity kinetic energy (TIKE) inlet is defined with the intensity value of 0.05 for both `inlet1` and `inlet2`, and the inlet value for inlet outlet boundary condition is the same as the initial condition of the internal field. Turbulent mixing length frequency (TMLF) inlet boundary conditions for `inlet1` and `inlet2` for ω have the mixing length value of 0.00063 m and 0.01057 m, respectively. The inlet outlet boundary condition for ω also has the same value for the internal field as its inlet value. Furthermore, the wall-function parameters

for ω and ν_t in the boundary patch `wall` are given by $C_\mu = 0.09$, $\kappa = 0.41$, and $E = 9.8$.

Table 4.1: Initial and boundary conditions for the outlet simulation

	inlet1	inlet2	outlet	artificial walls	wall	IC
U	mapped	$(3.6, 0, 0) \text{ m/s}$	zero grad	slip	no slip	$(3.6, 0, 0) \text{ m/s}$
T	mapped	1836.3 K	zero grad	zero grad	zero grad	$1.836 \times 10^3 \text{ K}$
p	zero grad	zero grad	total pressure	zero grad	zero grad	101325 Pa
α_t	mapped	calculated	calculated	wall function	wall func	0 kg/ms
k	TIKE inlet	TIKE inlet	inlet outlet	zero grad	wall func	$0.0486 \text{ m}^2/\text{s}^2$
ω	TMLF inlet	TMLF inlet	inlet outlet	zero grad	wall func	38.079 1/s
ν_t	mapped	calculated	zero grad	zero grad	wall func	$0 \text{ m}^2/\text{s}$
H_2	mapped	0	zero grad	zero gradient	zero grad	0
O_2	mapped	0.0915	zero grad	zero grad	zero grad	0.0915
N_2	mapped	0.7537	zero grad	zero grad	zero grad	0.7537
H_2O	0	0.1537	zero grad	zero grad	zero grad	0.1537
default	0	0	zero grad	zero grad	zero grad	0

Although the inlet simulation is performed using OpenFOAM v2312, the outlet simulation is run with OpenFOAM 11. To launch the case, the generic `foamRun` command is used instead of calling a solver explicitly. `foamRun` automatically detects the appropriate solver based on the definition given in the file `system/controlDict`. In this configuration, `foamRun` selects `multicomponentFluid`, which is the correct solver for multi-species reactive flows with detailed chemistry.

The outlet simulation is advanced in pseudo time from iteration 0 to iteration 8000. Field data are written to disk every 250 pseudo time steps, providing sufficient temporal resolution for monitoring convergence and flame behaviour. The last two pseudo time steps are present at the end of the simulation, from which the last pseudo time step is the converged solution.



(a) without inner mesh overlay



(b) with inner mesh overlay

Figure 4.2: `wall` patch

Parallelisation is achieved by decomposing the computational domain into 96 subdomains, corresponding to 96 CPU cores running simultaneously on a single node. The chosen decomposition method is the `simple` scheme. In this scheme, the mesh is divided into a structured grid of subdomains using three integers that specify the number of partitions in each spatial direction. Here, the domain is split into 8 divisions in the x -direction, 12 in the y -direction, and 1 in the out-of-plane direction, which is consistent with the two-dimensional nature of the outlet simulation geometry. The simulation runs on a single shared-memory filesystem rather than multiple distributed file systems.

For the discretisation and solver configuration of the outlet simulation; a first-order implicit time discretisation, upwind-stabilised convection schemes, linear gradient and Laplacian operators, and consistent interpolation and wall-distance formulations are employed. Linear systems for pressure, velocity, enthalpy, turbulence quantities, density, and species transport are solved using

a combination of PCG, GAMG, and smooth solvers, while the PIMPLE algorithm controls the iterative coupling with a single outer corrector and two inner correctors. The Courant number is restricted to ensure numerical stability. For clarity and reproducibility, the full `fvSchemes` and `fvSolution` dictionaries, as well as all solver tolerances and algorithmic parameters, are provided in Appendix A.

To introduce hydrogen combustion and NO_x chemistry, the `chemkinToFoam` utility is used. This tool converts standard `chemkin`-formatted mechanism files into inputs that are readable by OpenFOAM. The conversion is executed with the command:

```
chemkinToFoam chemkin/H2-NOX.chmech \  
chemkin/H2-NOX.chthermo \  
chemkin/transportProperties \  
constant/reactions \  
constant/thermo.compressibleGas
```

Here, the `.chmech` file contains the chemical reaction mechanism, i.e. all elementary reactions, Arrhenius coefficients, third-body efficiencies, fall-off formulations and reaction types. After conversion, the resulting OpenFOAM-formatted reactions appear in `constant/reactions`. The present finite-rate chemistry mechanism includes hydrogen oxidation reactions, chain-branching and termination reactions, the extended Zeldovich mechanism, the NNH pathway, N_2O -related NO production, NO/ NO_2 interconversion reactions. These pathways are essential for predicting NO and NO_2 levels and flame stability in lean or partially premixed hydrogen flames.

The `.chthermo` file provides NASA seven-coefficient polynomials for species thermodynamic properties. After conversion, these polynomials are written into `constant/thermo.compressibleGas`. This file contains the full list of 19 species, NASA polynomial coefficients for each species, transport parameters used by OpenFOAM.

The `transportProperties` file contains species transport data, e.g. the parameters of the chosen transport model.

The `chemkin` input files used in this work originate from the C3MechV4.0 [37] hydrogen- NO_x mechanism. Before conversion, their filenames were adapted as follows:

- C3MechLite.MECH $\rightarrow$ H2-NOX.chmech,
- C3MechLite.THERM $\rightarrow$ H2-NOX.chthermo,
- C3MechLite.TRAN $\rightarrow$ transportProperties.

Since OpenFOAM requires a different structure, the transport file `transportProperties` was created manually in OpenFOAM syntax based on the Chemkin `.TRAN` data, the other files are left untouched other than their names.

A frequent issue arises from strict formatting rules in the NASA polynomial sections. The third line of each NASA entry must follow the fixed-width format `3F10.0`, meaning three floating-point values, each occupying ten character columns. If this condition is violated, `chemkinToFoam` will fail with errors referencing specific line numbers. This issue is resolved by reformatting the affected sections of the `.chthermo` file before conversion.

4.2 Optimization Details

The goal of the optimization step is to identify inlet geometries that balance two competing performance criteria: (i) minimizing NO_x emissions and (ii) reducing the risk of flashback.

These two quantities inherently compete with one another, such that improvements in one direction frequently lead to degradations in the other. Therefore, rather than prescribing a single optimal design, the problem is formulated as a multi-objective optimization in which both quantities are minimized simultaneously as objective functions. The result is a Pareto front from which the designer can later select the preferred compromise based on application-specific trade-offs.

A variety of potential formulations are considered for the NO_x metric, for example peak local NO_x mass fraction, to avoid local hotspots, and volume-integrated NO_x production. Ultimately, the most physically meaningful and robust choice is found to be the total NO_x mass flow leaving the outlet domain,

$$J_{\text{NO}_x} = \dot{m}_{\text{NO}_x, \text{out}} = \int_{\Omega_{\text{outlet}}} \rho (Y_{\text{NO}} + Y_{\text{NO}_2}) \mathbf{u} \cdot \mathbf{n} \, dA, \quad (4.1)$$

which measures the transport of nitrogen oxides out of the system and directly reflects the environmental impact of the burner configuration. Y_{NO} and Y_{NO_2} are the mass fractions of NO and NO_2 , respectively. The integral is over the outlet patch Ω_{outlet} , where the total NO_x mass fraction is scaled by the density ρ , velocity in outwards normal direction $\mathbf{u} \cdot \mathbf{n}$, and the infinitesimal area dA .

A consistent discrete counterpart of this continuous definition, which is suitable for the 2D case analyzed in this thesis, is as follows:

$$J_{\text{NO}_x}^{\text{discrete}} = \dot{m}_{\text{NO}_x, \text{out}} = \sum_{e \in \Gamma_{\text{outlet}}} \rho_e (Y_{\text{NO}, e} + Y_{\text{NO}_2, e}) (\mathbf{u}_e \cdot \mathbf{n}_e) \ell_e. \quad (4.2)$$

Here, the outlet is given as the line Γ_{outlet} . The sum runs over all outlet boundary edges/segments e , ℓ_e is the segment length, and $\mathbf{u}_e \cdot \mathbf{n}_e$ is the velocity in outwards direction in 2D.

The chosen objective function for the flashback risk is the maximum value of the flashback indicator within a predefined region of interest, inspired by the flashback detection mechanisms introduced in Chapter 3:

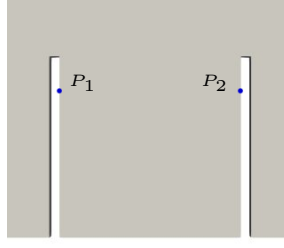
$$J_{\text{FB}} = \max_{\Omega_{\text{FB}}} \frac{g_c}{g}, \quad (4.3)$$

where g is the flow velocity gradient and g_c is the critical gradient.

Because the flashback analysis in this work is performed on a two-dimensional outlet domain, the region of interest Ω_{FB} reduces to two discrete wall points, P_1 and P_2 . These points must lie on the inner walls of the boundary patch `wall`, and the line connecting them must be oriented normal to the parallel walls of the domain. In a fully three-dimensional configuration, by contrast, the corresponding region of interest would form a one-dimensional curve that encircles the two-dimensional inlet boundary (e.g. `inlet1`) while lying on the two-dimensional boundary manifold `wall`, assuming a geometry analogous to the 2D case considered in this thesis.

In the implementation developed for this thesis, the two evaluation points P_1 and P_2 are chosen at 80 % of the `wall` height, depicted in Figure 4.3. The reason is to ensure that only points located sufficiently downstream, but not directly on the outlet of the patch `wall` are considered. These points are pre-determined to be $P_1 = (0.00725, 0.0045, 0.00025)$ and $P_2 = (0.00725, -0.0045, 0.00025)$, then hard coded to the post-processing script of the outlet simulation.

Because of the 2D nature of the outlet simulation, the flashback objective function becomes:

Figure 4.3: **wall** patch, the two dots representing P_1 and P_2

$$J_{\text{FB}}^{\text{discrete}} = \max_{i=1,2} \frac{g_c}{g_i}, \quad (4.4)$$

with g_i evaluated at the wall points P_1 and P_2 .

The value for the critical gradient g_c is approximated globally, and not in dependence of space, since the T-CVG Criterion [32] is utilized, as described in Chapter 3. The criterion states:

$$g_c = \frac{S_T}{Le_{\text{fuel}} \delta_p}. \quad (4.5)$$

The approximation of the turbulent flame speed used in this work is [38]:

$$S_T = 1 + 0.446 \left(\frac{Ze}{Pe} \right)^{0.450} \left(\frac{u'}{s_l} \right)^{0.607} \left(\frac{l_t}{\delta_l} \right)^{0.206}, \quad (4.6)$$

where Ze is the Zeldovich number, Pe is the Peclet number, s_l is the laminar flame speed, δ_l is the laminar flame thickness, u' is the velocity fluctuation, which is approximated as 10% of the area weighted average velocity magnitude in **inlet1** of the outlet simulation:

$$u' = \frac{0.1}{L_{\text{inlet1}}} \sum_{e \in \Gamma_{\text{inlet1}}} |\mathbf{u}_e| \ell_e. \quad (4.7)$$

The total boundary length L_{inlet1} is defined as:

$$L_{\text{inlet1}} = \sum_{e \in \Gamma_{\text{inlet1}}} \ell_e.$$

The variable l_t is the integral length scale that is approximated with the diameter, i.e. the distance between the inner walls of the boundary patch **wall**, as depicted in Figure 4.4. The value for l_t for the outlet simulation mesh used in this thesis is fixed at 0.009 m.

The values of S_T , Le_{fuel} and δ_p are dependent on the equivalence ratio, which is given as 0.6 in the inlet simulation. Therefore, the data for the variables Ze , Pe , s_l , δ_t , δ_p and Le_{fuel} are calculated with the kinetics solver FlameMaster [39] and a helper Python tool [40], and are shown in Table 4.2.

The script approximates the flow velocity gradient $\partial u / \partial y$ at the two wall locations P_1 and P_2 using a first-order forward finite-difference formula. For each evaluation point P_i , the method uses

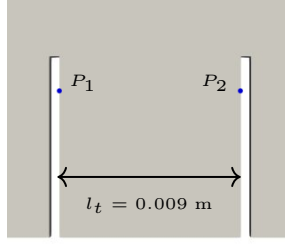


Figure 4.4: The length measurement between the inlet walls

Table 4.2: Values of the variables in the T-CVG Criterion for premixed hydrogen flame at $\phi = 0.6$

Ze	7.87
Pe	0.589
s_l	0.915 m/s
δ_l	0.00036 m
δ_p	0.00036 m
Le_{fuel}	0.28

$$u_{\text{wall}}(P_i), \quad u_{\text{int}}(P_i), \quad \Delta y_i,$$

where $u_{\text{wall}}(P_i)$ is the velocity sampled on the `wall` patch, $u_{\text{int}}(P_i)$ is the velocity at the nearest interior cell centre, and Δy_i is the wall-normal displacement between these two sampling locations. Formally,

$$\Delta y_i = y_{\text{int}}(P_i) - y_{\text{wall}}(P_i), \quad (4.8)$$

and the forward finite-difference approximation of the wall-normal gradient is

$$\left. \frac{\partial u}{\partial y} \right|_{P_i} \approx \frac{u_{\text{int}}(P_i) - u_{\text{wall}}(P_i)}{\Delta y_i}. \quad (4.9)$$

The implementation then considers the magnitude of this gradient

$$g_i = \left| \left. \frac{\partial u}{\partial y} \right|_{P_i} \right|. \quad i = 1, 2. \quad (4.10)$$

The choice for the flashback objective function is motivated by the fact that flashback is a strictly local phenomenon: even a small region exhibiting high flashback propensity can destabilize the flame. Minimizing the maximum value therefore ensures that all the flashback ratios in the chosen region are minimized, and provides a conservative yet informative measure of flashback risk that is independent of global averaging effects, such that the designer then can choose the design parameters that yield a flashback ratio of less than 1.

An alternative formulation considered during this work was to treat the flashback safety requirement as a hard constraint. However, imposing a strict constraint of the form

$$J_{\text{FB}} < 1 \quad (4.11)$$

is generally too restrictive for practical optimization. In practice, several softer formulations exist. One option is to introduce a safety margin and require

$$J_{\text{FB}} < 1 - \varepsilon, \quad (4.12)$$

or, in a Bayesian optimization setting, to enforce a probabilistic constraint of the form

$$\mathbb{P}(J_{\text{FB}} < 1 - \varepsilon) \geq p. \quad (4.13)$$

These relaxed approaches reduce the risk of overly conservative designs and place more emphasis on minimizing NO_x , which is often the more important objective under already safe operating conditions. Nevertheless, using constraints restricts the set of design points available to the designer. For multi-objective studies, it is often preferable to keep both NO_x and flashback risk as explicit objectives, enabling the user to select a preferred design from the full Pareto-optimal set rather than from a prematurely filtered solution space. For this reason, the final implementation adopts two explicit objectives without hard constraints.

For debugging and testing purposes, a further objective function for the absolute pressure loss between the inlet and outlet is defined. For each boundary, an area-averaged static pressure is computed, and solutions that introduce excessive pressure losses between inlet and outlet are penalized, which ensures maintaining a physically consistent flow behavior. The difference is measured between the inlet, i.e. the transverse fuel inlets and the main oxidizer inlet, of the non-reactive simulation and the outlet of the reactive simulation, where the spatial average is taken. In the continuous limit, the objective function is written as:

$$J_p = |\Delta p| = \left| \frac{1}{A_{\text{out}}} \int_{\Omega_{\text{out}}} p(\mathbf{x}) \, dA - \frac{1}{A_{\text{in}}} \int_{\Omega_{\text{in}}} p(\mathbf{x}) \, dA \right|, \quad (4.14)$$

where A_{out} and A_{in} are defined as the outlet and inlet area, respectively.

Because the outlet case is 2D, the version used in this work is defined as the difference between the scaled sums over the outlet line and the inlet plane, where ℓ_e is the length of the segment e , A_e is the area of the plane e , Γ_{out} is the outlet line and Ω_{in} is the total inlet plane.

$$J_p^{\text{discrete}} = |\Delta p| = \left| \frac{1}{L_{\text{out}}} \sum_{e \in \Gamma_{\text{out}}} p_e \ell_e - \frac{1}{A_{\text{in}}} \sum_{e \in \Omega_{\text{in}}} p_e A_e \right|, \quad (4.15)$$

with the total boundary lengths L_{out} and A_{in} defined as:

$$L_{\text{out}} = \sum_{e \in \Gamma_{\text{out}}} \ell_e, \quad A_{\text{in}} = \sum_{e \in \Omega_{\text{in}}} A_e.$$

The optimization is performed using the Dakota framework, which interacts with the geometry generation software Salome through an `apptainer` image (`salome.sif`). Geometry updates are executed via the Python script `L50624_IGNITER_GeomExport.py`, which parametrically modifies the DAP inlet according to the design variables proposed by Dakota. The optimization setup is defined in a dedicated Dakota input file that specifies the variables, the evaluation interface, the simulation driver script, maximum number of iterations and the optimization method.

During the development phase, the single-objective surrogate-based optimizer EGO was first tested on simplified configurations. Surrogate-based methods require an initial set of sampling points, which were generated using Latin Hypercube Sampling. A separate Dakota input file was used to generate these samples.

To reduce compute cost during development, a synthetic “dummy” evaluation script was temporarily employed to generate artificial data for testing the surrogate construction and EGO loop. A realistic setup would require on the order of 100–120 high-fidelity CFD samples, which is prohibitively expensive for initial prototyping. Also, surrogate-based Gaussian process

models are ill-suited for discrete design variables such as the number of inlets $N_{\text{fuel_inlets}}$, which prevents including such variables in the Gaussian process kernel. Furthermore, inside the Dakota framework, surrogate modeling approaches are not implemented for multi-objective optimization problems. Because of these limitations, EGO was deemed unsuitable for the final multi-objective problem.

The final optimization approach employs Dakota’s MOGA method, a multi-objective genetic algorithm. Dakota’s MOGA is simpler than state-of-the-art algorithms such as NSGA-II, however it implements the essential evolutionary concepts required to approximate the Pareto front.

Although MOGA is significantly more computationally expensive than EGO, requiring many direct CFD evaluations and providing fewer samples for a given computational budget, it is well suited for the present problem because it naturally handles multi-objective formulations, does not require smoothness or differentiability of the objective functions, and thus accommodates discrete variables without modification.

For the implementation, the existing Dakota file was adapted to include the MOGA-specific parameters, and the sampling was eliminated from the process. Also, the post-processing scripts are modified to output two objective values in `results.out`. Lastly, a further post-processing script is included, such that after every optimization cycle, the nondominated designs are extracted and the corresponding Pareto front is constructed. This final set of solutions represents the optimal trade-off between minimizing NO_x and flashback risk.

The specific operator settings used for the genetic algorithm are chosen to balance exploration of the design space with convergence toward Pareto-optimal solutions. These settings are documented in Table 4.3.

Parameter	Value	Explanation
Population size	4	Each generation contains four design candidates.
Crossover type	<code>shuffle_random</code>	Offsprings are generated by randomly shuffling and recombining parent variables.
Mutation type	<code>replace_uniform</code>	Mutation replaces selected design variables with uniformly sampled values within their bounds.
Fitness type	<code>domination_count</code>	Fitness based on the number of designs dominating a candidate (lower is better).
Seed	81075	Ensures deterministic reproducibility.

Table 4.3: MOGA configuration parameters used in the optimization

4.3 The Proposed Optimization Workflow

The resulting optimization workflow developed in this thesis builds upon the methodology described in the reference paper, but extends it by introducing a second, fully reactive outlet simulation. This additional stage enables the evaluation of NO_x emissions and flashback risk, which were not accessible in the original non-reactive framework.

To maintain modularity and clarity, the implementation is separated into two main directories, which are also depicted in Figure 4.5:

- **Dakota folder**, containing the Dakota input files, geometry generation scripts, SLURM submission file, and the inlet (non-reacting) simulation,

- **reactive simulation folder**, containing the reactive outlet case, its SLURM submission file and its post-processing scripts.

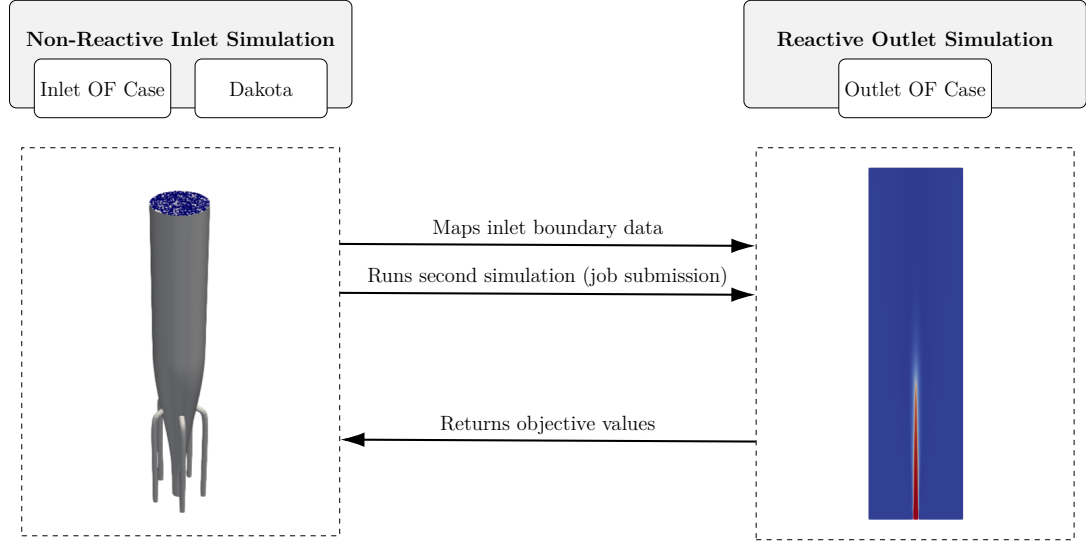


Figure 4.5: Folder structure of the optimization workflow

The Dakota SLURM job file `slurm.dakota` resides in the Dakota directory and launches the entire evaluation pipeline. The job uses `#SBATCH --ntasks=96`, matching the domain decomposition chosen for the CFD simulations.

The environment dependencies for the Dakota job are loaded as:

```

module load GCC/13.3.0
module load OpenMPI/5.0.3
module load Dakota/6.22.0
ml foss/2024a OpenFOAM/v2312

```

During the optimization, Dakota creates a sequence of evaluation folders (`ego_run1`, `ego_run2`, ...), each folder containing an isolated copy of the inlet simulation case and all scripts needed for that evaluation step.

For every evaluation, the following operations are executed:

1. **Detecting the evaluation number.** Each `ego_runN` folder first extracts N from its name, which allows the post-processing scripts to identify the correct output directory for copying results, and helps bookkeeping.
2. **Duplicating the outlet simulation case.** This creates a sequence of outlet evaluation folders (`run1`, `run2`, ...), analogous to the inlet simulation workflow. Here, the evaluation number extracted in the previous step is used to determine the folder name.
3. **Pre-processing of the inlet simulation.** Geometry generation is performed through the `Salome.sif` container, using the Python script `L50624.IGNITER.GeomExport.py`. Mesh generation follows.
4. **Execution of the inlet simulation.** The non-reacting RANS inlet case is run in parallel on 96 subdomains.

5. **Post-processing and data transfer to the outlet simulation.** Species mass fractions (H_2 , O_2 , N_2), turbulence variables (k , ω , α_t , ν_t), velocity U , and temperature T are sampled at the outlet patch of the inlet simulation. These sampled fields are radially averaged, mirrored, interpolated onto the inlet boundary of the outlet case, and written as initial and boundary conditions for the outlet case. Since the coordinate systems of the inlet and outlet simulations differ, i.e. the inlet domain is rotated by 90 degrees relative to the outlet domain, the x- and y-components of the velocity field U must be interchanged during the mapping process. A schematic illustration of this mapping procedure is provided in Fig. 4.6.

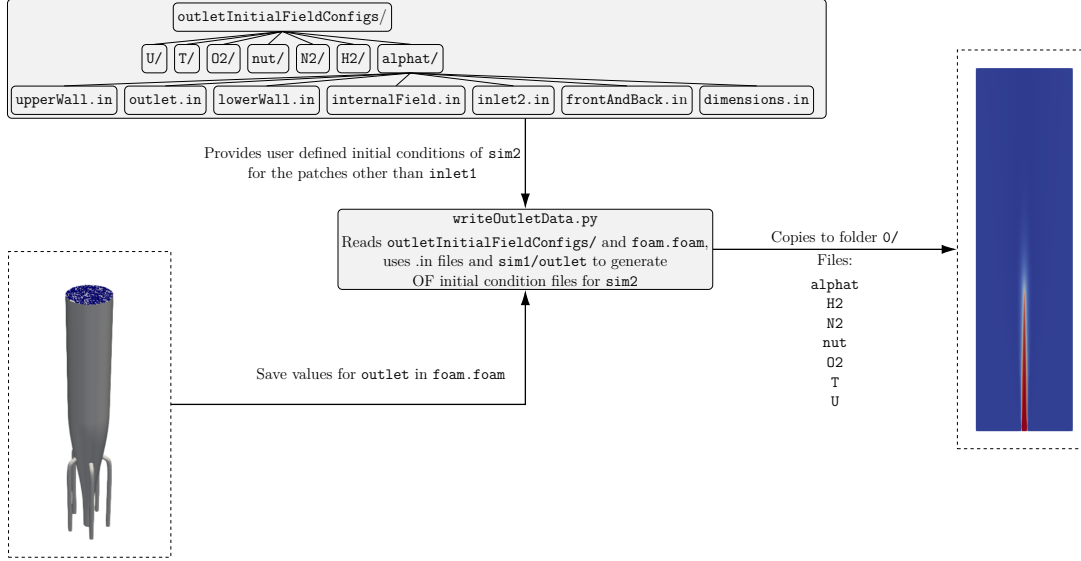


Figure 4.6: Schematic of the data mapping workflow from the 3D inlet simulation to the 2D reactive outlet simulation

The remaining initial fields of the outlet simulation (e.g. zero-gradient fields, wall fields) are stored in the folder `outletInitialFieldConfigs`, which is read during preprocessing to create the initial condition files that are copied to the outlet simulation case. The initial condition files of the properties that are not mapped from the non-reactive inlet simulation are left untouched.

6. **Launching the reactive outlet simulation.** After preparing the boundary and initial conditions, the case is launched in the current evaluation folder `runN` as a child SLURM process by the script `RunSimulationOutlet.sh`. The main Dakota job waits until the child job finishes. The outlet simulation uses a different module environment:

```

module load GCC/11.3.0
module load OpenMPI/4.1.4
module load OpenFOAM/11
  
```

7. **Post-processing of the outlet simulation.** Once the simulation has converged, two key performance indicators are extracted:

- **NO_x mass flow.** The script integrates the species mass fractions of NO and NO_2 at the outlet to compute the total mass flow of nitrogen oxides, see Equation 4.1.
- **Flashback risk metric.** The flashback indicator field is evaluated over the user-defined flashback domain. The algorithm extracts the maximum flashback indicator

in that region, see Equation 4.4.

Because the simulation is steady-state RANS, time-dependent upstream flame propagation cannot be captured. Likewise, combustion-instability-induced flashback and CIVB mechanisms are not represented, as the configuration contains no swirl and no unsteady structures.

Additionally, mainly for testing purposes, **pressure loss** is implemented as a further performance indicator. The communication between the two post-processing scripts of the simulation cases are done via the file `p_inlet.out` that is copied and pasted from the non-reactive case to the post-processing folder of the reactive case.

8. **Writing the results.** The two objective values (J_{NO_x} and J_{FB} , or J_{NO_x} and J_p) are written to `results.out` and moved back to the corresponding `ego_runN` folder.
9. **Appending to the database and plotting Pareto front.** The Python script `plot_pareto.py` automatically appends to the internal database `output.dat`, which has the same structure as the Dakota-generated database `table_out.dat`, and plots the final Pareto front of optimal inlet geometries by reading the internal database. The labels of the plot are read from the file `parameter_names.out`, which is created by the post-processing script of the outlet simulation, and copied into the Dakota folder.
10. **Dakota reads the objective values.** Dakota updates the optimization state by writing to `table_out.dat` and initiates the next evaluation. In MOGA, the exact number of total evaluation runs is not given at the beginning of the optimization loop.

A directed graph of the complete final CFD-Dakota optimization workflow is depicted in Figure 4.7.

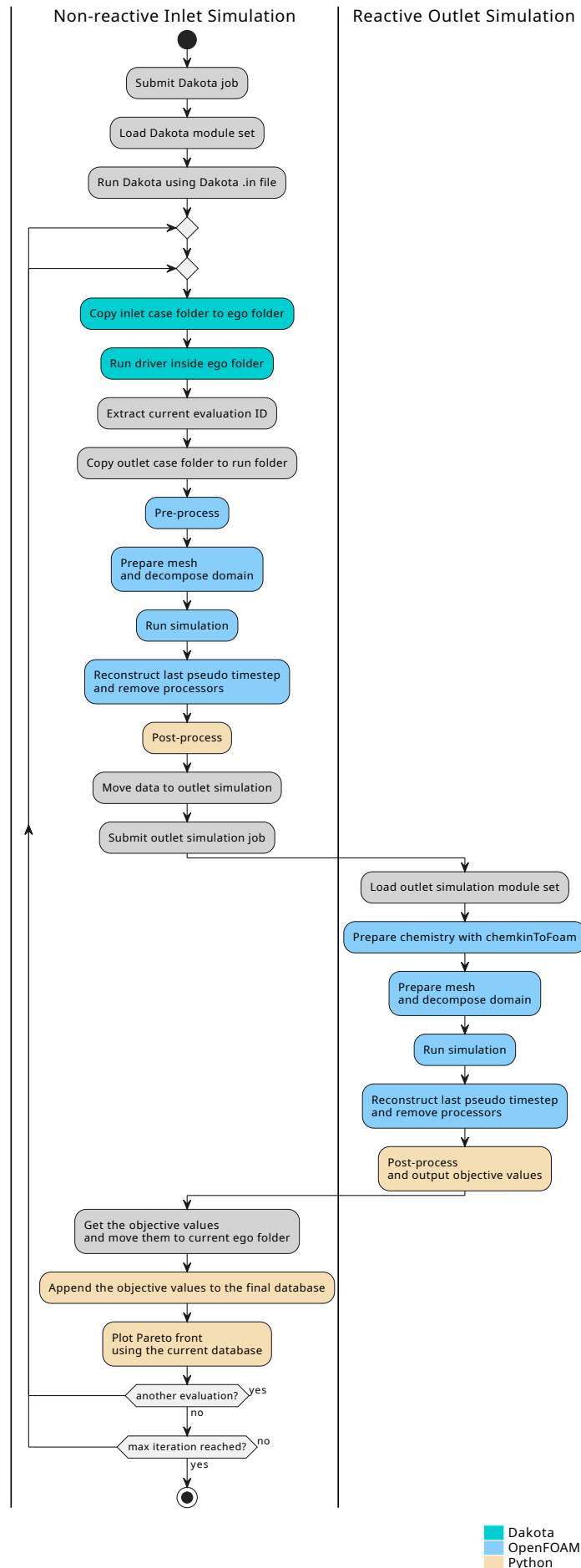


Figure 4.7: Activity diagram of the complete CFD-Dakota optimization workflow

Chapter 5

Results and Discussion

This chapter presents the results obtained from applying the developed CFD-based multi-objective optimization workflow to a parametrized hydrogen burner inlet geometry. The workflow couples steady-state RANS simulations with automated geometry generation and multi-objective evolutionary optimization in order to systematically explore the trade-off between flashback safety and NO_x emissions. A two-stage simulation strategy is employed, consisting of a non-reactive inlet simulation to resolve the upstream mixing field and a reactive outlet simulation to evaluate combustion behavior, thermal NO_x formation, and flashback risk. Quantitative objective functions are defined for both performance criteria: the total NO_x mass flow at the outlet and a conservative flashback risk indicator based on near-wall velocity gradients. Additionally, pressure loss is implemented as an objective function, mainly for debugging and testing purposes. By embedding these metrics directly into the optimization loop, the methodology transforms CFD from a purely diagnostic tool into a design-driving component that enables reproducible and automated assessment of competing combustion objectives.

The focus of this chapter lies on analyzing the performance of different optimization strategies and on interpreting the resulting trade-offs between NO_x emissions, flashback risk, and pressure loss. Two optimization cases are investigated, from which all are minimization cases: (i) a MOGA optimization considering NO_x emissions and pressure loss, and (ii) a MOGA optimization considering NO_x emissions and flashback risk. For each case, both global objective trends and local flow and combustion characteristics are analyzed using post-processed simulation data.

For the four design parameters α , h , d and $N_{\text{fuel.inlets}}$, the same lower and upper bounds stated in the baseline work [6] are used, as given previously in Table 3.2. The initial and boundary conditions for the reactive outlet simulation are taken as given in Chapter 4. Since the baseline work [6] does not state initial and boundary conditions for all the field variables in the non-reactive inlet simulation, these are chosen independently of the baseline work, and presented in Table 5.1.

For k , turbulent intensity kinetic energy (TIKE) inlet is defined with the intensity value of 0.0471 for the oxidizer inlet and 0.0628 for the fuel inlet. Turbulent mixing length dissipation rate (TMLDR) inlet boundary conditions for the oxidizer inlet for ω have the mixing length value of 0.0025 m and for the fuel inlet 0.000038 m. The mass flow rate for the flow rate inlet velocity (FRIV) boundary condition in the oxidizer inlet is set as 0.000724356 kg/s and in the fuel inlet as 0.00001274309 kg/s. Furthermore, analogous to the reactive outlet simulation case, the wall-function parameters for ϵ are given by $C_\mu = 0.09$, $\kappa = 0.41$, and $E = 9.8$. Also as in the outlet case, the turbulent Prandtl number is chosen as 0.85.

Table 5.1: Boundary conditions for the boundary patches of the inlet simulation

	walls	oxidizer inlet	fuel inlet	outlet	IC
U	no slip	FRIV	FRIV	zero grad	$(35.141, 0, 0) \text{ m/s}$
T	zero grad	298.15 K	298.15 K	zero grad	298.15 K
p	zero grad	zero grad	zero grad	total pressure	101325 Pa
α_t	wall func	calculated	calculated	calculated	0 kg/ms
k	wall func	TIKE inlet	TIKE inlet	zero gradient	$56.728 \text{ m}^2/\text{s}^2$
ϵ	wall func	TMLDR inlet	TMLDR inlet	zero gradient	$15381.691 \text{ m}^2/\text{s}^3$
ν_t	wall func	calculated	calculated	calculated	$0 \text{ m}^2/\text{s}$
H_2	zero grad	0	1	zero grad	0
O_2	zero grad	0.23	0	zero grad	0.23
N_2	zero grad	0.77	0	zero grad	0.77
default	zero grad	0	0	zero grad	0

5.1 Test Analyses

To gain physical insight into the optimization process and the simulation cases, one design evaluation run is analyzed further in the following. The design parameters of this case are given in Table 5.2.

Table 5.2: Design parameters for the analyzed evaluation run

α [°]	h [mm]	d [mm]	$N_{\text{fuel_inlets}}$
59.803591701	17.462609120	1.158601443	4.785342105 (rounded to 5)

Since the sampling algorithm for the design parameters assumes continuous variables, the discrete design variable $N_{\text{fuel_inlets}}$ is rounded to the nearest integer after the sampling step. This rounding operation introduces discontinuities into the design space, which leads to unstable behavior when using MOGA. In general, surrogate-based optimization approaches are better suited for problems involving discrete design variables. Nevertheless, for the sake of simplicity, MOGA is retained in the present work for the optimization loop, while allowing the number of fuel inlets to vary as a design parameter.

After the end of the non-reactive inlet simulation, the variables α_t , H_2 , N_2 , ν_t , O_2 , T and U are mapped as an 18-point profile to the reactive outlet simulation, as depicted previously in Figures 4.1 and 4.6. The results of the non-reactive inlet simulation run for a vertical slice of the 3D burner domain are shown in Figure 5.1. The 1D mapped profiles and the corresponding 2D distributions of these variables at the outlet of the non-reactive simulation are shown in Figure 5.2.

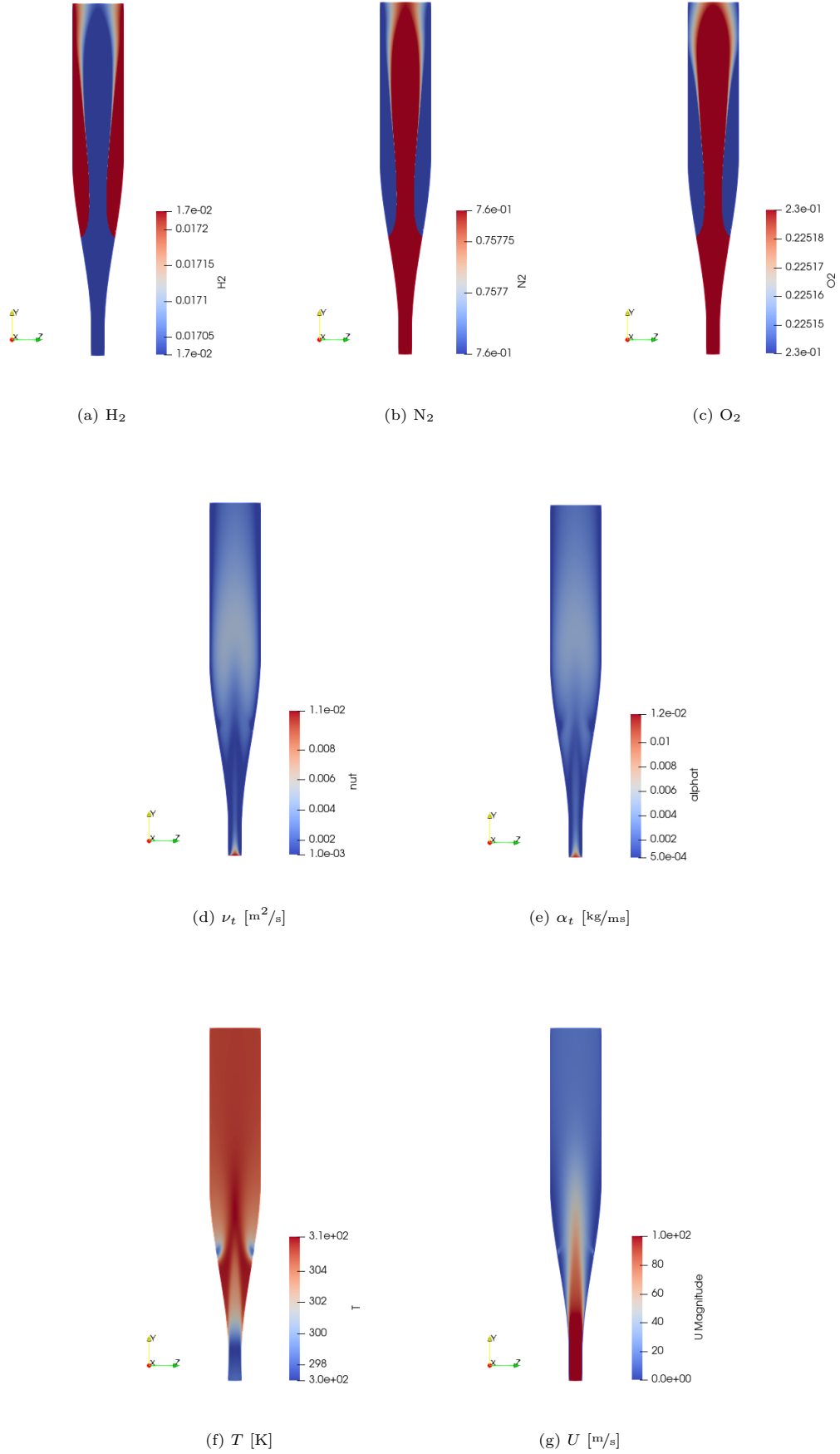


Figure 5.1: RANS simulation results for a vertical slice of the 3D burner domain

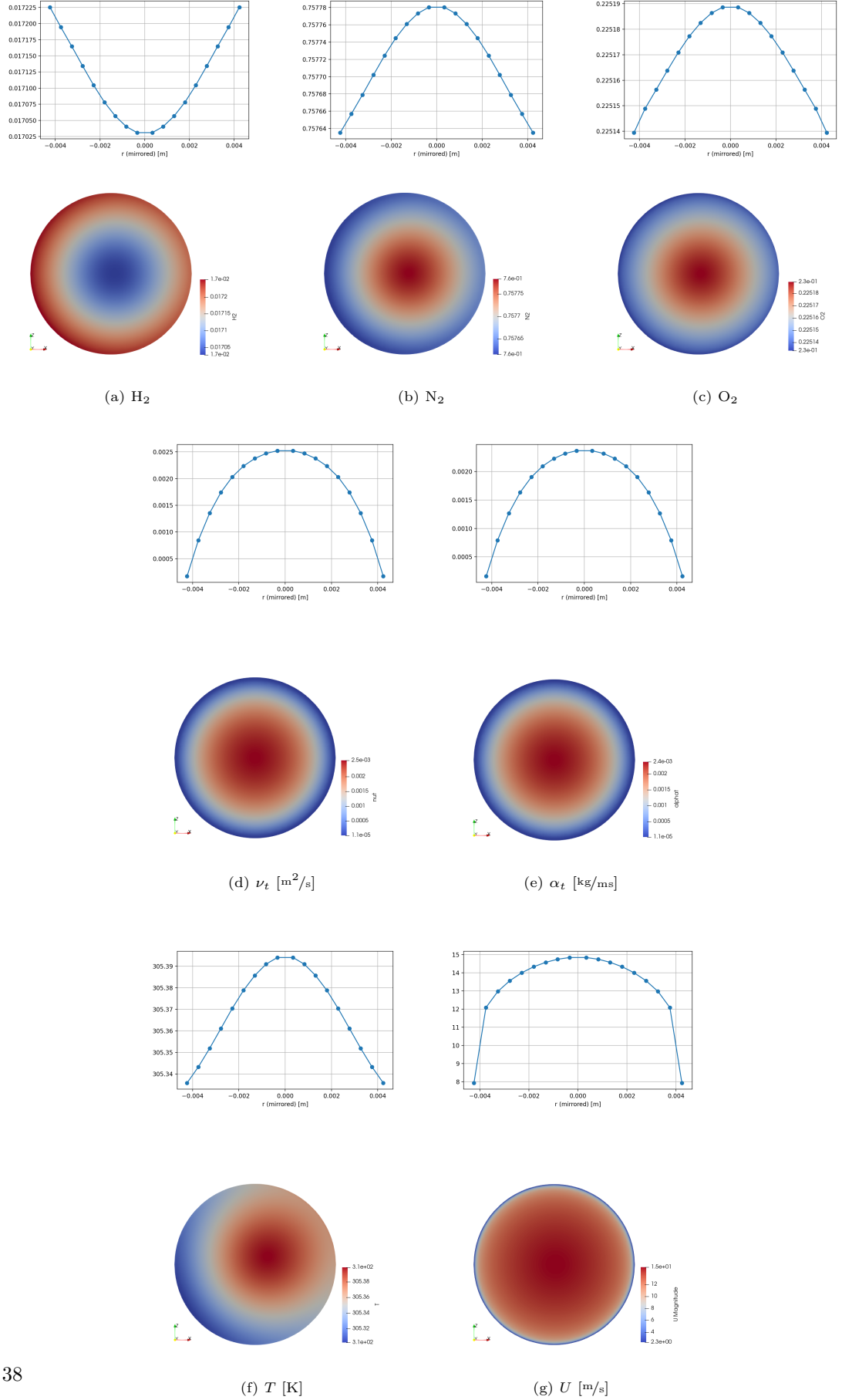


Figure 5.2: Distributions of mapped inlet quantities used as boundary conditions for the reactive outlet simulation

All mapped inlet profiles exhibit smooth, approximately axisymmetric radial distributions with pronounced extrema at the centerline. The turbulent quantities α_t and ν_t , as well as the velocity magnitude U , temperature T , mass fractions of O_2 and N_2 display maxima near the centerline, while the mass fraction of H_2 depicts a deficit there. Overall, the smooth and symmetric profiles indicate a physically consistent mapping procedure and provide well-defined inlet conditions for the subsequent reactive simulation.

The increase in velocity U at the centerline reflects a high-momentum region, surrounded by a slower annular flow. The velocity distribution is a consequence of the jet-in-crossflow (JICF) burner configuration, where the transverse injection of fuel (H_2) into a main oxidizer stream (O_2, N_2) leads to the formation of a concentrated core jet that persists toward the outlet. The elevated centerline velocity suggests that axial momentum is predominantly transported in the core region, while the surrounding flow experiences enhanced mixing and momentum diffusion due to stronger shear layers. Turbulence production is highest in the middle outer regions in the downstream of the injection, where velocity gradients are largest, while the centerline is characterized by reduced shear and a more coherent flow structure. Furthermore, the flow experiences a downstream deceleration, which is primarily a result of turbulent momentum mixing and redirection of axial momentum. Overall, the mapped inlet profiles represent a physically plausible fuel-rich jet surrounded by an oxidizer-rich coflow, which constitutes a realistic and well-defined initial condition for the reactive simulations.

Within the employed k - ϵ turbulence closure, the turbulent kinematic viscosity ν_t is computed from the turbulent kinetic energy k and its dissipation rate ϵ that influence each other with the model's transport equations:

$$\nu_t = C_\mu \frac{k^2}{\epsilon},$$

where the model constant C_μ is given. Moreover, the turbulent thermal diffusivity α_t is directly related to ν_t through the turbulent Prandtl number,

$$Pr_t = \frac{\nu_t}{\alpha_t} \Rightarrow \alpha_t = \frac{\nu_t}{Pr_t},$$

such that assuming $Pr_t = \text{const.}$ (here $Pr_t = 0.85$) implies $\alpha_t \propto \nu_t$ and explains why both quantities exhibit qualitatively similar radial distributions, including the centerline maxima.

The mapped values are then used as inlet boundary conditions in the reactive outlet simulation. The results for the RANS reactive outlet simulation are shown in Figure 5.3.

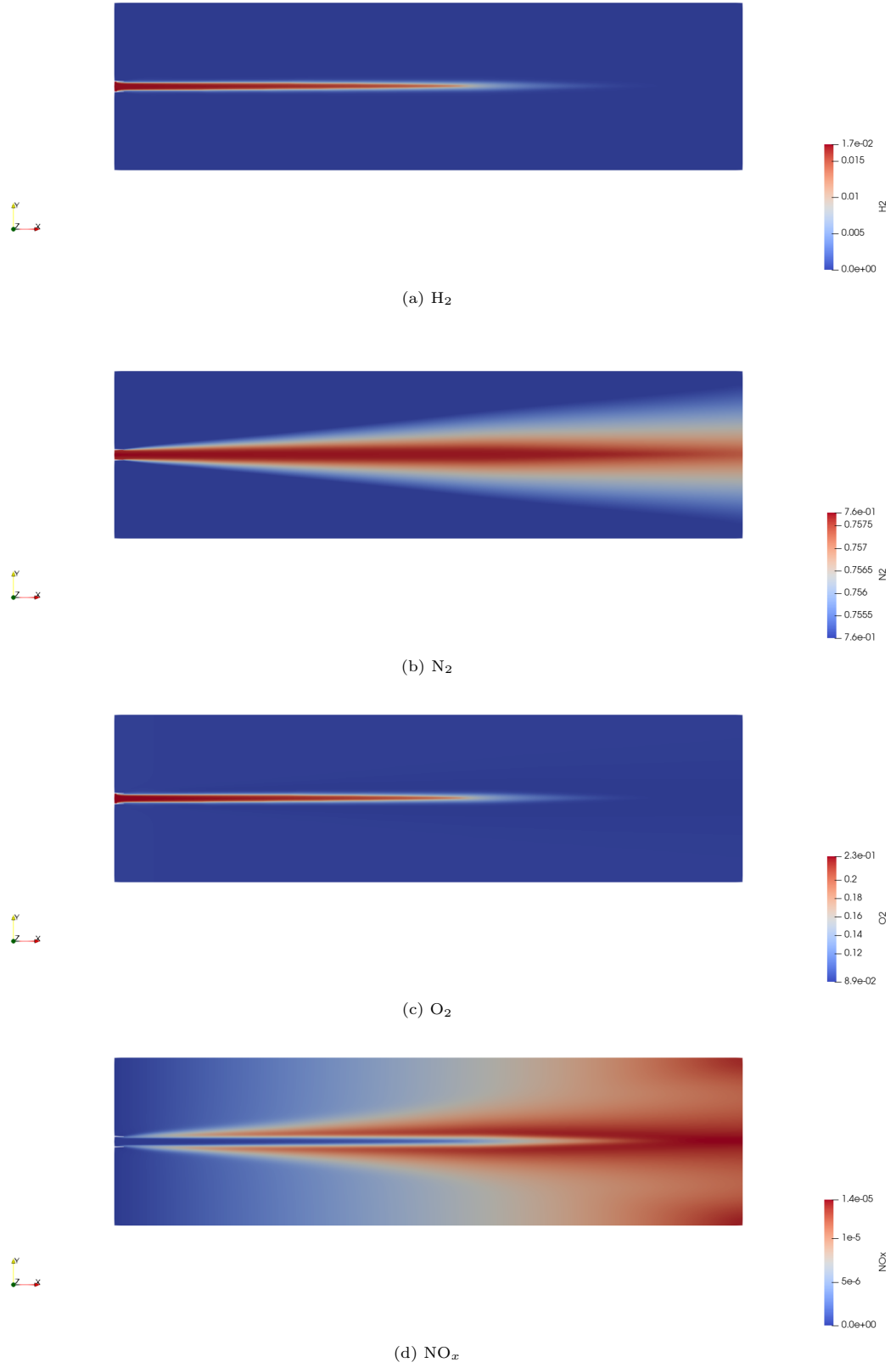


Figure 5.3: The results for the RANS outlet simulation

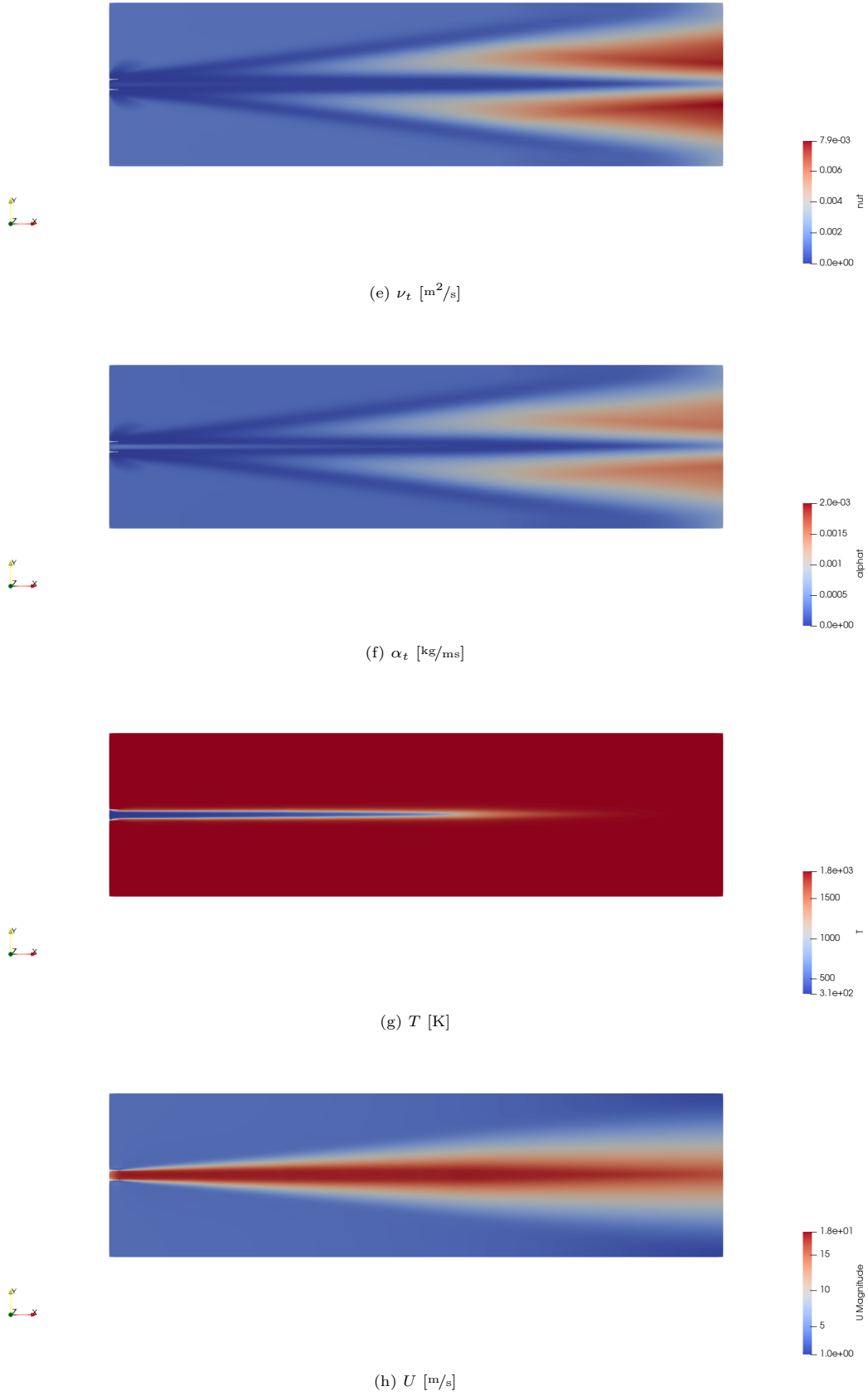


Figure 5.3: The results for the RANS outlet simulation (continued)

Figure 5.3 illustrates the mean-field structure of the reactive outlet simulation and shows how the stratified inlet flow evolves under the combined effects of turbulent mixing and chemical conversion. The hydrogen field is initially confined to a thin inlet core and decays downstream due to consumption, while oxygen demonstrates a very similar behavior. Nitrogen becomes increasingly dominant toward the outlet as reactants are consumed and products accumulate. The NO_x mass fraction remains negligible close to the inlet and increases progressively downstream, reflecting finite-rate thermal NO_x formation that depends on residence time and elevated post-flame temperatures. The turbulent transport fields ν_t and α_t exhibit pronounced maxima in the shear layers and grow downstream, demonstrating strong turbulence production driven by velocity and scalar gradients. The temperature field transitions from a colder inlet region to a largely elevated level downstream, with a comparatively cooler centerline, while the axial velocity magnitude shows a spreading high-speed core whose peak decays downstream as axial momentum is redistributed by turbulent mixing.

After completion of the reactive outlet simulation, the resulting objective value is obtained as $J_{\text{NO}_x} = 2.63 \times 10^{-6} \text{ kg/s}$. As seen in Figure 5.4, the radial distribution of the NO_x mass fraction exhibits a pronounced maximum in the central region; however, the profile is not strictly monotonic. Following a local minimum at intermediate radii, the NO_x level increases again toward the outer region of the outlet. In contrast, the temperature distribution shows a broad central plateau with two slightly elevated off-center maxima and a shallow dip at the centerline, followed by a gradual decrease toward the outer radii.

As a consequence, the observed NO_x distribution at the outlet cannot be attributed solely to the instantaneous local temperature field. Instead, it reflects the cumulative NO_x formation history along the flow path, which is governed not only by the strong temperature sensitivity of thermal NO formation via the Zeldovich mechanism, but also by local mixture composition, residence time, and convective–turbulent transport effects. While elevated NO_x levels remain partially correlated with the generally hotter central flow region, the non-monotonic behavior of the radial profile indicates that transport and mixing processes play a significant role in shaping the outlet NO_x distribution beyond a purely temperature-driven mechanism.

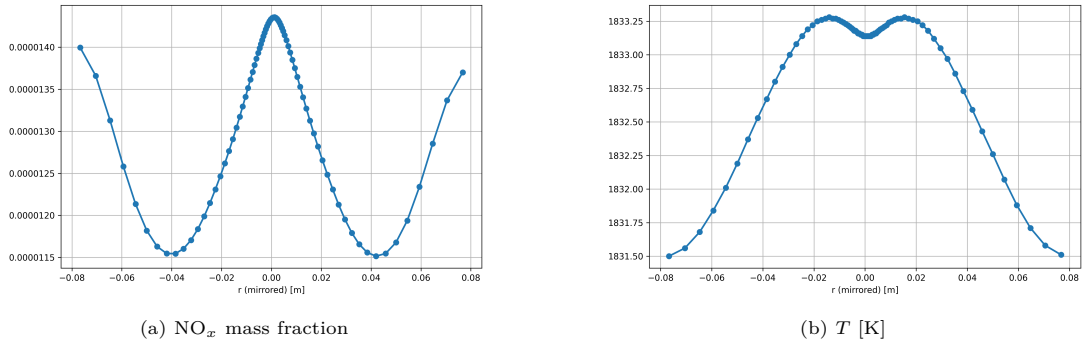


Figure 5.4: NO_x fraction and temperature distributions in the reactive outlet simulation for the test evaluation case

5.2 NO_x-Pressure Loss Optimization

A preliminary optimization study was conducted using a multi-objective genetic algorithm, considering NO_x emissions and pressure loss as competing objectives. The total wall-clock time of the optimization run amounted to 1 day and 9 hours, executed on a single compute node. The maximum number of optimization iterations was limited to two because of computational limits and the lack of a restarting mechanism for the optimization. The evaluation concurrency was set to one, which resulted in a total of ten individual design evaluations.

Considering the complete optimization process rather than a single generation, the resulting Pareto front for the NO_x-pressure loss optimization is presented in Figure 5.5. The corresponding design parameter sets evaluated during the optimization are summarized in Table 5.3, where the Pareto-optimal points are highlighted in bold font. In Figure 5.6, the parameter data points are plotted vertically and scaled, such that the lower limit and upper limit, as defined in Figure 3.2, are scaled to 0 and 1, respectively.

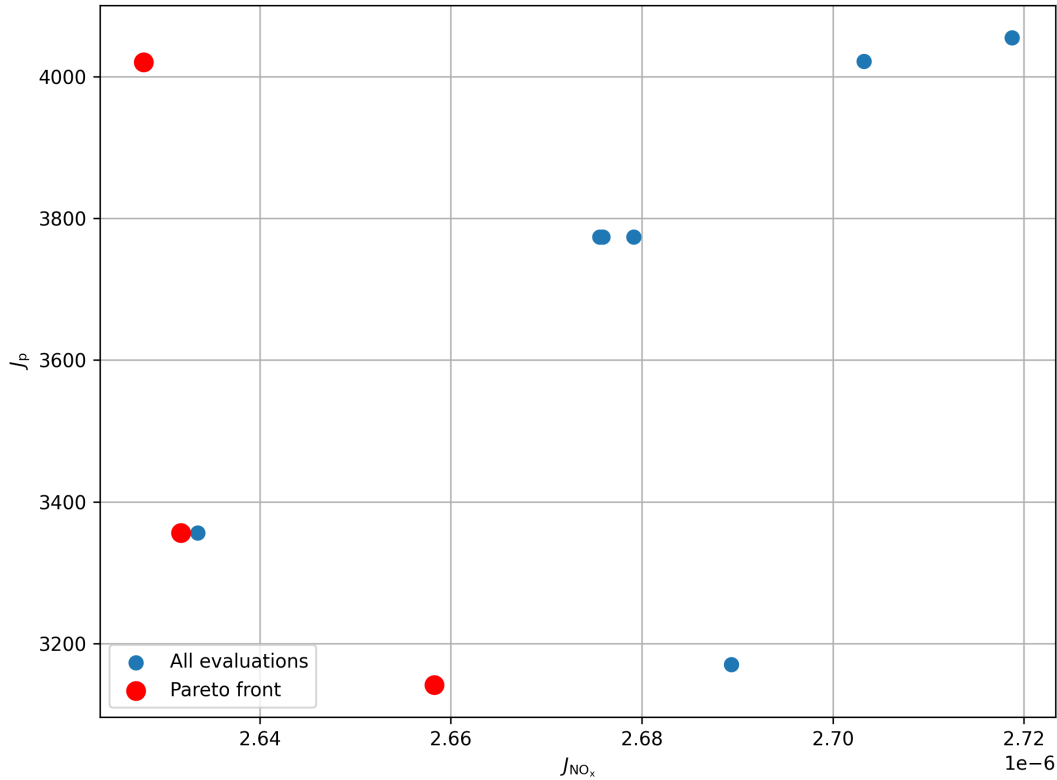
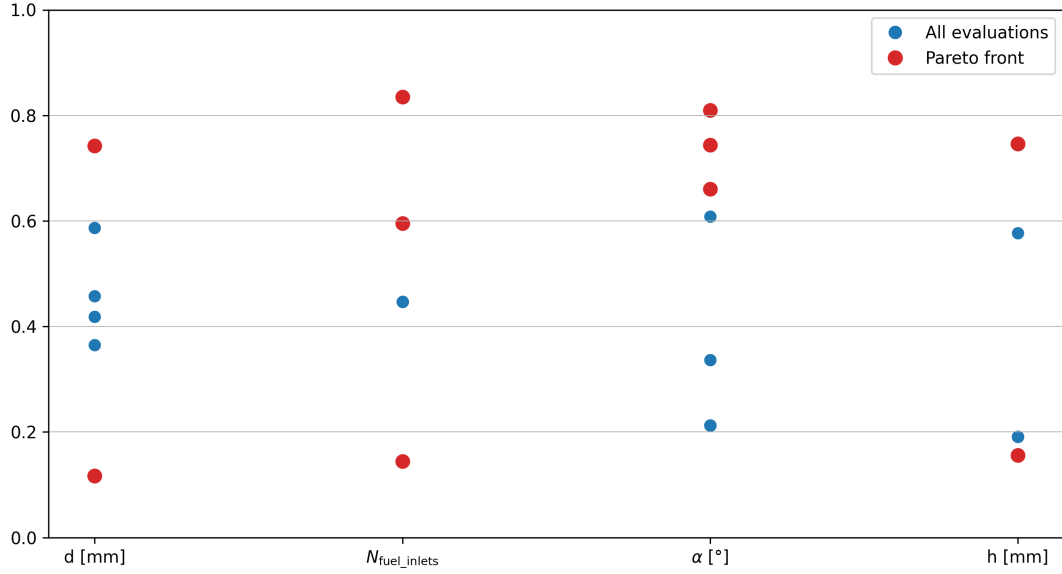


Figure 5.5: Pareto front of NO_x mass flow [kg/s] versus pressure loss [Pa] obtained from the MOGA optimization

Table 5.3: Design parameters with J_{NO_x} and J_p values for the evaluated configurations

Evaluation ID	d [mm]	$N_{\text{fuel_inlets}}$	α [°]	h [mm]	J_{NO_x} [kg/s]	J_p [Pa]
1	1.1586	4.7853	59.8036	17.4626	2.6317×10^{-6}	3356.23
2	1.2823	5.5044	62.3254	15.7674	2.6894×10^{-6}	3170.28
3	1.3933	4.3397	46.3796	11.9146	2.6759×10^{-6}	3774.02
4	1.4711	3.4341	64.2931	11.5565	2.6278×10^{-6}	4020.70
5	1.1586	4.3397	46.3796	11.9146	2.6756×10^{-6}	3774.18
6	1.1586	4.7853	58.2622	17.4626	2.6335×10^{-6}	3356.65
7	1.1586	5.5044	62.3254	17.4626	2.6583×10^{-6}	3141.57
8	1.3093	4.3397	46.3796	11.9146	2.6791×10^{-6}	3774.08
9	1.3289	3.4341	64.2931	11.5565	2.7032×10^{-6}	4022.30
10	1.4711	3.4341	50.0872	11.5565	2.7188×10^{-6}	4055.27

Figure 5.6: Normalized parameter distributions for J_{NO_x} - J_p optimization

The results of the multi-objective optimization demonstrate the trade-off between NO_x emissions and pressure loss. The distribution of evaluated designs (blue markers) illustrates a wide exploration of the objective space, while the Pareto front (red markers) defines the Pareto-optimal data points.

Three distinct optimal configurations are identified on the frontier. The first solution minimizes J_{NO_x} to $2.6278 \times 10^{-6} \text{ kg/s}$ but incurs a higher pressure penalty with $J_p = 4010.70 \text{ Pa}$. In contrast, a further solution prioritizes pneumatic efficiency, reducing J_p to 3141.57 Pa at the cost of increased emissions with $J_{\text{NO}_x} = 2.6583 \times 10^{-6} \text{ kg/s}$. The last solution is a trade-off in the middle of pneumatic efficiency and small NO_x emissions with the objective values $J_{\text{NO}_x} = 2.6317 \times 10^{-6} \text{ kg/s}$ and $J_p = 3356.23 \text{ Pa}$.

Figure 5.6 shows that the complete set of evaluated designs spans a wide range of normalized values for each design parameter, while the Pareto-optimal solutions are, for the most part, distributed across unconfined regions of the design space rather than clustering around distinct

extrema.

The injector diameter d exhibits a pronounced spread among Pareto-optimal solutions, with both lower and higher normalized values represented. Qualitatively, smaller diameters are associated with lower pressure loss but increased NO_x emissions, whereas larger diameters tend to increase pressure loss while reducing NO_x. However, in combination with suitably chosen values of the remaining design parameters, smaller diameters can still yield configurations with only moderately increased pressure loss and moderately elevated NO_x emissions. This indicates that reduced injector diameters can constitute a favorable compromise for simultaneously limiting NO_x formation and pressure loss when the overall inlet geometry is optimally configured.

A similarly broad distribution is observed for the number of fuel inlets $N_{\text{fuel.inlets}}$ and the inlet height h , both of which show Pareto-optimal solutions at lower as well as higher normalized values. In contrast to the diameter, the inlet height exhibits a reversed qualitative trend, where larger values of h may lead to moderately increased pressure losses while maintaining moderately elevated NO_x emissions. This suggests that increased inlet heights can represent a viable compromise solution, provided that the remaining parameters are adjusted accordingly.

For the three Pareto-optimal designs, the number of fuel inlets corresponds to rounded integer values of 3, 5, and 6, indicating that the explored design space is qualitatively well covered, and a trend towards optimal number of fuel inlets cannot be extracted. It should be noted here, that $N_{\text{fuel.inlets}}$ is treated as a continuous variable in the present optimization, despite being inherently discrete from both physical and manufacturing perspectives.

Finally, the injection angle α shows a clear preference for intermediate to higher normalized values among the Pareto-optimal solutions. This indicates that larger injection angles promote enhanced mixing in a manner that is simultaneously beneficial for NO_x reduction and pressure-loss minimization.

These results establish a quantitative foundation for design selection, where the final configuration must be negotiated between the three Pareto-optimal points based on the prioritized weighting of environmental impact versus aerodynamic efficiency. However, the current Pareto front exhibits a sparse distribution, with only three out of ten data points being Pareto-optimal, suggesting that the optimization has not yet converged to a clear Pareto front. By increasing the number of generations beyond the current limit of two, a smoother and more continuous frontier could be resolved. Furthermore, extended optimization would allow the genetic algorithm to explore previously unvisited regions of the design space, e.g. with crossover and mutation mechanisms of MOGA. With only two generations, the algorithm is largely restricted to the neighborhood of the initial stochastic sampling. By increasing the number of optimization steps, crossover can effectively recombine high-performing traits from disparate individuals, while mutation introduces random variations that prevent the population from stagnating in local optima. This continuous injection of genetic diversity allows the algorithm to probe the unvisited regions of the objective space. As the population evolves, the selection pressure iteratively discards dominated solutions, pushing the entire population toward the true global Pareto front, which is a process that requires sufficient generational cycles to allow for the emergence and refinement of superior design candidates.

5.3 NO_x-Flashback Risk Optimization

Following the preliminary testing of the optimization framework, the multi-objective optimization was executed to address the primary research objective of this thesis: the simultaneous minimization of NO_x emissions and flashback risk. The total wall-clock time of the optimization run amounted to 1 day and 10 hours, also executed on a single compute node, while maintaining the same restrictive optimization parameters used in previous cases: a maximum of two generations and a concurrency level of one. This configuration resulted in a total of ten design evaluations. Similar to the analysis of the pressure loss trade-off, the identified Pareto-optimal set for emissions and flashback risk is illustrated in Figure 5.7. The specific design parameters corresponding to these solutions are summarized in Table 5.4, where the Pareto-optimal points are again highlighted in bold font, and plotted in Figure 5.8.

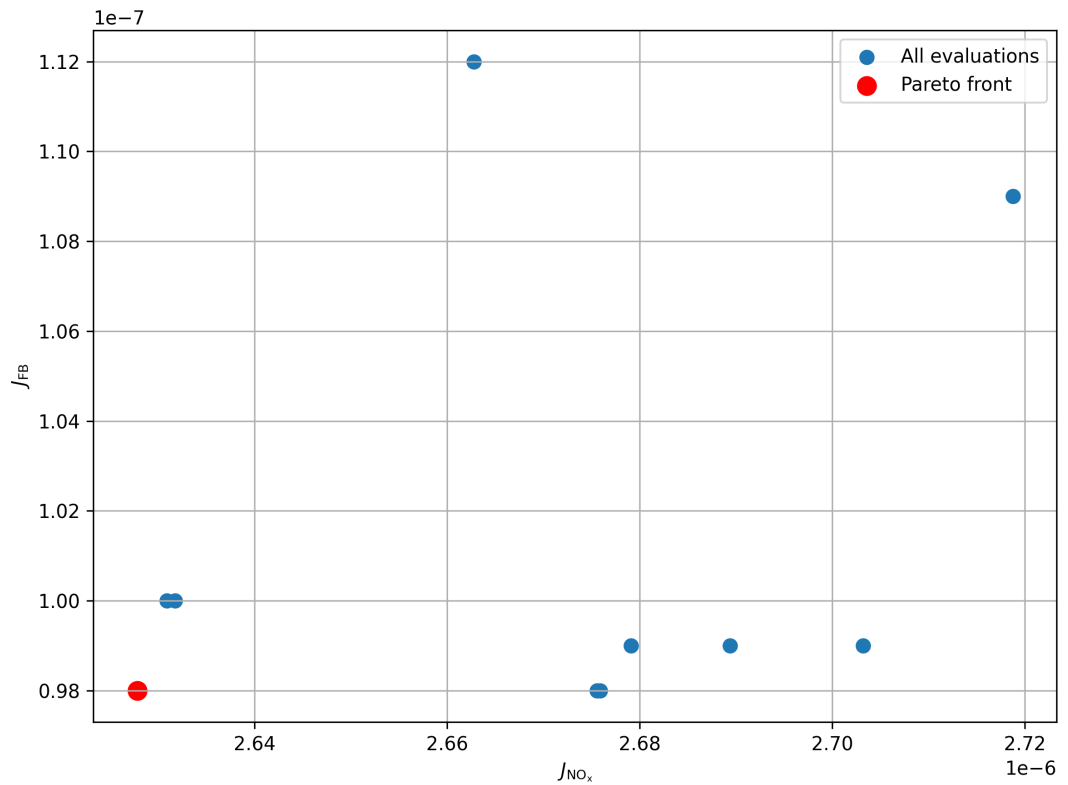
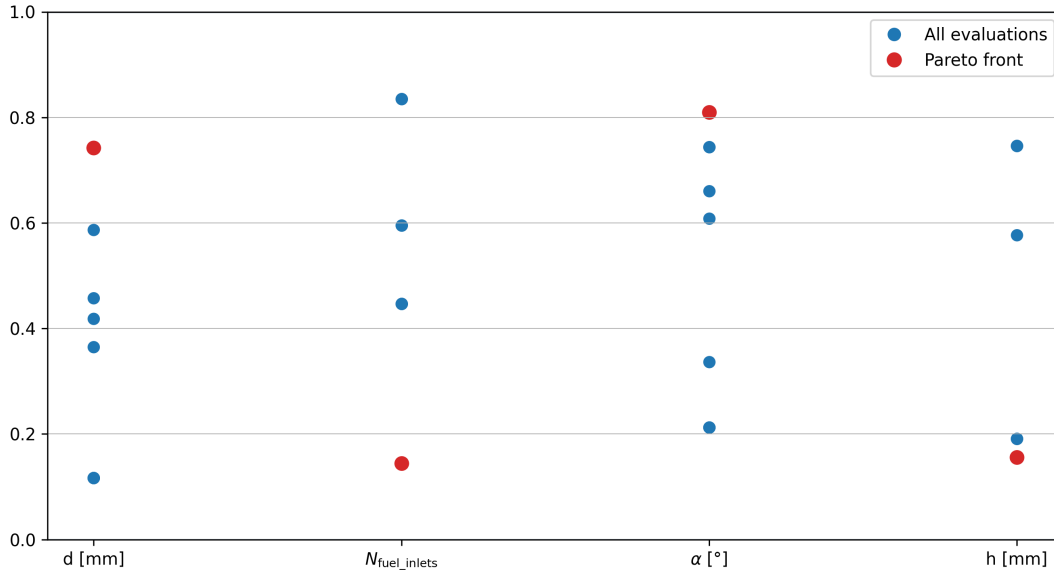


Figure 5.7: Pareto front of NO_x mass flow [kg/s] versus flashback risk obtained from the MOGA optimization

Table 5.4: Design parameters with J_{NO_x} and J_{FB} values for the evaluated configurations

Evaluation ID	d [mm]	$N_{\text{fuel_inlets}}$	α [°]	h [mm]	J_{NO_x} [kg/s]	J_{FB}
1	1.1586	4.7853	59.8036	17.4626	2.6317×10^{-6}	1.00×10^{-7}
2	1.2823	5.5044	62.3254	15.7674	2.6894×10^{-6}	0.99×10^{-7}
3	1.3933	4.3397	46.3796	11.9146	2.6759×10^{-6}	0.98×10^{-7}
4	1.4711	3.4341	64.2931	11.5565	2.6278×10^{-6}	0.98×10^{-7}
5	1.1586	4.3397	46.3796	11.9146	2.6756×10^{-6}	0.98×10^{-7}
6	1.1586	4.7853	58.2622	17.4626	2.6335×10^{-6}	1.00×10^{-7}
7	1.1586	5.5044	62.3254	17.4626	2.6583×10^{-6}	1.12×10^{-7}
8	1.3093	4.3397	46.3796	11.9146	2.6791×10^{-6}	0.99×10^{-7}
9	1.3289	3.4341	64.2931	11.5565	2.7032×10^{-6}	0.99×10^{-7}
10	1.4711	3.4341	50.0872	11.5565	2.7188×10^{-6}	1.09×10^{-7}

Figure 5.8: Normalized parameter distributions for J_{NO_x} - J_{FB} optimization

In contrast to the previously discussed NO_x-pressure loss optimization, the present objective space exhibits only a weak sensitivity with respect to the flashback risk metric. Across all evaluated design configurations, the flashback objective remains confined to a narrow interval between $J_{\text{FB}} = 0.98 \times 10^{-7}$ and $J_{\text{FB}} = 1.12 \times 10^{-7}$. Also, the Pareto front collapses to a single point (Evaluation ID 4), which simultaneously corresponds to the global minimum in NO_x emissions ($J_{\text{NO}_x} = 2.6278 \times 10^{-6}$ kg/s) and one of the three configurations exhibiting the lowest flashback risk ($J_{\text{FB}} = 0.98 \times 10^{-7}$). Although the absolute variation in J_{FB} is small, these differences are fully resolved at double precision and remain meaningful when compared to the characteristic magnitude of the NO_x objective, which is approximately one order of magnitude larger. This indicates that, despite the narrow range, relative comparisons of flashback risk between designs remain physically interpretable.

The current optimization budget of only two generations is insufficient to induce pronounced variations in the flashback risk metric or to activate potential trade-offs between NO_x emissions and flashback risk. To resolve a meaningful Pareto front and to explore possible coupling effects

between these objectives, analogous to those observed in the NO_x -pressure loss optimization, a substantially increased number of optimization iterations would be required. This would enable the genetic algorithm to systematically explore regions of the design space where flashback-related flow and mixing characteristics vary more strongly.

The resulting Pareto plot may however also be attributed to the limited influence of the selected design parameters on the objective functions, leading to insufficient sensitivity within the explored design space. If the Pareto front does not converge in a reasonable manner, even after more than two optimization iterations, this indicates that the current parametrization may be inadequate to capture the governing physics of flashback. In such a case, alternative design parameters that more directly affect flashback propensity should be identified and incorporated into the optimization framework.

As shown in Figure 5.8, the single Pareto-optimal design is characterized by the largest injector diameter, the smallest number of fuel inlets, the largest injection angle, and the smallest inlet height among the evaluated configurations. Nevertheless, given the collapse of the Pareto front to a single solution, these observed trends should be interpreted as indicative rather than definitive. A conclusive assessment of qualitative parameter dependencies would require a significantly expanded optimization campaign with increased population size and number of generations.

To further analyze the flashback risk, the velocity gradients and fluctuations are evaluated at the predefined wall points. The results summarized in Table 5.5 provide a detailed breakdown of the components contributing to the flashback objective function J_{FB} .

Table 5.5: Flashback-related quantities for all design configurations in the optimization run

Evaluation ID	$ \partial u / \partial y _{P_1}$ [1/s]	$ \partial u / \partial y _{P_2}$ [1/s]	u' [m/s]	$J_{\text{FB}} = \max_{\Omega_{\text{FB}}} \frac{g_c}{g}$
1	5.7833×10^4	5.7898×10^4	1.3229	1.00×10^{-7}
2	5.8261×10^4	5.9536×10^4	1.3222	0.99×10^{-7}
3	5.8602×10^4	5.8559×10^4	1.3215	0.98×10^{-7}
4	5.8533×10^4	5.8456×10^4	1.3210	0.98×10^{-7}
5	5.8604×10^4	5.8562×10^4	1.3215	0.98×10^{-7}
6	5.7801×10^4	5.7870×10^4	1.3228	1.00×10^{-7}
7	5.7170×10^4	5.1498×10^4	1.3229	1.12×10^{-7}
8	5.8100×10^4	5.7989×10^4	1.3215	0.99×10^{-7}
9	5.8200×10^4	5.8854×10^4	1.3212	0.99×10^{-7}
10	5.2623×10^4	5.9514×10^4	1.3215	1.09×10^{-7}

As defined in Equation 4.4, J_{FB} is determined by the maximum ratio of the critical velocity gradient g_c to the local velocity gradient g_i at the predefined wall points P_1 and P_2 .

An analysis of the numerical values reveals that the velocity fluctuations u' remain nearly constant across all ten evaluations, with a minimal variance around a value of 1.32 m/s . Since u' is a primary driver for the turbulent flame speed S_T and the subsequent critical gradient g_c (Equations 4.5 and 4.6), the numerator of the objective function remains largely invariant. However, the local velocity gradients $|\partial u / \partial y|$ at points P_1 and P_2 show considerable sensitivity to the changes in design parameters d , $N_{\text{fuel.inlets}}$, α , and h , maintaining magnitudes between $5.1498 \times 10^4 \text{ 1/s}$ and $5.9536 \times 10^4 \text{ 1/s}$. Consequently, the resulting J_{FB} values exhibit low, but non-negligible variance.

Chapter 6

Conclusion and Outlook

This thesis developed and demonstrated a workflow that integrates RANS simulations into an optimization framework for identifying inlet geometries that balance flashback risk and NO_x emissions. The developed CFD-based workflow achieves the primary goal of this thesis by establishing a modular and automated framework for the multi-objective optimization of hydrogen burner geometries. The methodology is intended as a preliminary proof-of-concept that defines the current boundaries of simulation-driven design, although having several limiting factors that influenced the optimization results presented in this work. By identifying these specific limitations, this work sets the foundation for future research, from which subsequent developments can build upon to improve accuracy and efficiency. The limitations and possible future research regarding filling these gaps are explained in this chapter.

A central conclusion of this work is that surrogate-based approaches are highly promising for the optimization of complex combustion systems, as they offer an effective trade-off between computational cost and the theoretical ability to approximate global optima. Compared to gradient-based or purely genetic algorithms, surrogate-assisted methods are particularly attractive when the evaluation of the objective function relies on computationally expensive numerical simulations. Bayesian Optimization serves as a typical example of such surrogate-based modeling techniques and is commonly used in simulation-driven optimization tasks.

In the present study, however, the optimization relied on a multi-objective genetic algorithm (MOGA). This decision was primarily guided by two limitations: the absence of a surrogate-based multi-objective method in Dakota and the lack of an initial set of sampling data. Constructing a sufficiently large surrogate-training dataset would have required more than one hundred simulations for four design variables, making the approach computationally prohibitive within the scope of this thesis. Also, utilizing surrogate-based approaches may require reformulating the design space, for instance by removing the discrete inlet number $N_{\text{fuel.inlets}}$ as a variable.

An additional consideration highlighted by this work concerns the use of RANS simulations. While RANS provides an affordable modeling strategy, it has inherent weaknesses, particularly in flows characterized by large-scale mixing between regions of different temperature or composition. An avenue for improvement lies in the use of LES or multifidelity strategies instead of pure RANS to enhance physical fidelity. LES would generally resolve such dynamics more accurately, but its computational cost renders it unsuitable for direct integration within iterative optimization loops. A promising solution discussed in the literature, such as the approach presented in [23], is multifidelity RANS-LES optimization, where multi-objective methods are combined with hierarchical fidelity levels to improve predictive accuracy while containing computational demands.

Besides, the current methodology utilizes a detailed finite-rate chemical mechanism to solve

transport equations for all 19 species individually. While this approach provides high chemical resolution, it is computationally intensive for iterative optimization loops. Future work could improve both efficiency and physical accuracy by adopting flamelet models. These models are better tuned for hydrogen’s unique properties and can significantly reduce computational costs by utilizing pre-tabulated chemistry rather than resolving the production rates and transport for every species in every cell.

Additionally, the codebase can be improved from a software engineering perspective by enhancing scalability and modifiability. For example, a parameters file (e.g. `optimization_params.in`) can be implemented, such that the parameters for different types of optimization runs can be changed conveniently, in which they are automatically read and used by the scripts.

Furthermore, extending the optimization to additional objectives such as mixture uniformity or pressure loss would enable a more holistic evaluation of stability, emission reduction, and operational efficiency. Such extensions naturally lead to multi-dimensional (three or more objectives) optimization problems, in which the visualization and interpretation of Pareto sets becomes increasingly challenging.

Also, future research should extend the optimization beyond two generations to achieve a more converged and continuous Pareto frontier. While increasing generations would allow for better exploration of the design space via crossover and mutation, MOGA is computationally expensive because it requires numerous direct high-fidelity CFD evaluations. These computational demands can be managed by using more than one node in the high-performance computing cluster, as well as implementing a restarting mechanism for the optimization.

Another important open aspect of the present optimization framework concerns the normalization of the objective functions. While the employed multi-objective genetic algorithm is fundamentally based on Pareto dominance and therefore does not strictly require normalization, it remains unknown and untested whether an explicit scaling of the objectives would influence the convergence behavior, diversity of the Pareto front, or the distribution of solutions in objective space. In particular, the large differences in magnitude between NO_x emissions, flashback risk and pressure loss may affect secondary algorithmic mechanisms such as distance measures. Future work should therefore systematically assess the impact of objective normalization, for example by scaling NO_x with respect to the total mass flow or a reference NO_x mass flow value, expressing pressure loss as a dimensionless loss coefficient, and applying reference- or threshold-based scaling for the flashback risk indicator.

A flashback criterion other than T-CVG [32] may also be used to approximate the flashback risk, e.g. ET-CVG [17], since T-CVG has several downsides, when it comes to accuracy, as explained in Chapter 3. Especially the method of gradients is local, while the burning velocity is calculated globally, which leaves an area for improvement.

Lastly, advances in surrogate modeling open additional perspectives for future work. Methods such as response surfaces, support vector machines, neural networks, and modern machine-learning architectures are rapidly evolving and offer promising capabilities for design space exploration. Nevertheless, issues such as generalizability, integration into industrial workflows, and thorough validation including uncertainty quantification remain essential prerequisites for their practical adoption. Finally, the use of surrogate models in industrial contexts necessitates large and diverse training datasets. This raises important questions regarding the balance between synthetic CFD data and experimental measurements, which is an aspect closely connected to the development of digital twin technologies that may ultimately enable more robust and transferable data-driven combustion models.

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Appendix A

Numerical Settings

This appendix documents the numerical discretisation schemes and linear-solver settings used in the outlet simulation. The complete `fvSchemes` and `fvSolution` configurations are listed below for reproducibility.

A.1 `fvSchemes`

Time Discretisation

- Time derivative: `localEuler` (first-order, robust for reacting flows)

Gradient Schemes

- All gradients computed using `Gauss linear` (second-order)

Divergence Schemes

- Default divergence: `none`
- `div(phi,U)`: `Gauss upwind`
- `div(phiid,p)`: `Gauss upwind`
- `div(phi,K)`: `Gauss upwind`
- `div(phi,k)`, `div(phi,epsilon)`, `div(phi,omega)`: `Gauss upwind`
- `div(phi,Yi_h)`: `Gauss upwind`
- Viscous stress divergence `div(((rho*nuEff)*dev2(T(grad(U)))))` : `Gauss linear`

Laplacian Schemes

- All Laplacian terms use `Gauss linear uncorrected`

Interpolation Schemes

- Field interpolation: `linear`

Surface-Normal Gradient Schemes

- `snGrad`: `uncorrected`

Wall Distance Calculation

- `wallDist: meshWave`

A.2 fvSolution

Density Solvers (`rho.*`)

- Solver: `PCG`
- Preconditioner: `DIC`
- Solved to machine precision (no explicit tolerances)

Pressure Solvers (`p`, `pFinal`, `Phi`)

- Solver: `GAMG`
- Smoother: `DICGaussSeidel`
- Absolute tolerance: `1e-6`
- Relative tolerance: `0.05`

Other Fields (`U`, `h`, `e`, `k`, `omega`, `Yi`)

- Solver: `smoothSolver`
- Smoother: `GaussSeidel`
- Relative tolerance: `0.1`
- Maximum iterations: `20`

PIMPLE Algorithm

- Number of outer correctors: `1`
- Number of inner correctors: `2`
- Non-orthogonal correctors: `0`
- Momentum predictor: `enabled`
- Maximum Courant number: `0.5`
- Time-step smoothing coefficient: `0.5`
- Maximum time step: `1 s`

Relaxation Factors

- All equations solved with a relaxation factor of `1` (full relaxation)

Appendix B

Initial and Boundary Conditions

Here are the initial and boundary conditions of the non-reactive inlet and reactive outlet simulations are listed.

B.1 Inlet Simulation

Table B.1: Boundary conditions for the boundary patches of the inlet simulation

	walls	oxidizer inlet	fuel inlet	outlet	IC
U	no slip	FRIV	FRIV	zero grad	$(35.141, 0, 0) \text{ m/s}$
T	zero grad	298.15 K	298.15 K	zero grad	298.15 K
p	zero grad	zero grad	zero grad	total pressure	101325 Pa
α_t	wall func	calculated	calculated	calculated	0 kg/ms
k	wall func	TIKE inlet	TIKE inlet	zero gradient	$56.728 \text{ m}^2/\text{s}^2$
ϵ	wall func	TMLDR inlet	TMLDR inlet	zero gradient	$15381.691 \text{ m}^2/\text{s}^3$
ν_t	wall func	calculated	calculated	calculated	$0 \text{ m}^2/\text{s}$
H_2	zero grad	0	1	zero grad	0
O_2	zero grad	0.23	0	zero grad	0.23
N_2	zero grad	0.77	0	zero grad	0.77
default	zero grad	0	0	zero grad	0

- TIKE intensity (oxidizer) = 0.0471
- TIKE intensity (fuel) = 0.0628
- TMLDR mixing length (oxidizer) = 0.0025 m
- TMLDR mixing length (fuel) = 0.000038 m
- FRIV mass flow rate (oxidizer) = 0.000724356 kg/s
- FRIV mass flow rate (fuel) = $0.00001274309 \text{ kg/s}$
- $C_\mu = 0.09$
- $\kappa = 0.41$
- $E = 9.8$
- Turbulent Prandtl number = 0.85

B.2 Outlet Simulation

Table B.2: Initial and boundary conditions for the outlet simulation

	inlet1	inlet2	outlet	artificial walls	wall	IC
U	mapped	$(3.6, 0, 0) \text{ m/s}$	zero grad	slip	no slip	$(3.6, 0, 0) \text{ m/s}$
T	mapped	1836.3 K	zero grad	zero grad	zero grad	$1.836 \times 10^3 \text{ K}$
p	zero grad	zero grad	total pressure	zero grad	zero grad	101325 Pa
α_t	mapped	calculated	calculated	wall function	wall func	0 kg/ms
k	TIKE inlet	TIKE inlet	inlet outlet	zero grad	wall func	$0.0486 \text{ m}^2/\text{s}^2$
ω	TMLF inlet	TMLF inlet	inlet outlet	zero grad	wall func	38.079 1/s
ν_t	mapped	calculated	zero grad	zero grad	wall func	$0 \text{ m}^2/\text{s}$
H_2	mapped	0	zero grad	zero gradient	zero grad	0
O_2	mapped	0.0915	zero grad	zero grad	zero grad	0.0915
N_2	mapped	0.7537	zero grad	zero grad	zero grad	0.7537
H_2O	0	0.1537	zero grad	zero grad	zero grad	0.1537
default	0	0	zero grad	zero grad	zero grad	0

- TIKE intensity = 0.05
- TMLF mixing length (inlet1) = 0.00063 m
- TMLF mixing length (inlet2) = 0.01057 m
- $C_\mu = 0.09$
- $\kappa = 0.41$
- $E = 9.8$
- Turbulent Prandtl number = 0.85